AN AI ECOSYSTEM FOR FAKE JOB SCAM DETECTION AND FRAUD-AWARE JOB RECOMMENDATION

H. M. Nipuni Tharushika Bandara

(IT22251664)

BSc (Hons) degree in Information Technology, specializing in Software Engineering

Department of Software Engineering

Sri Lanka Institute of Information Technology
Sri Lanka

October 2026

AN AI ECOSYSTEM FOR FAKE JOB SCAM DETECTION AND FRAUD-AWARE JOB RECOMMENDATION

INDIVIDUAL THESIS

H. M. Nipuni Tharushika Bandara

(IT22251664)

Individual component: Employer Legitimacy Verification

Group ID: R26-SE-002

Report submitted in partial fulfilment of the requirements for the

BSc (Hons) degree in Information Technology

Specializing in Software Engineering

Department of Software Engineering

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

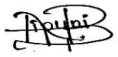
DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration. I declare that this thesis is my own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of my knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. I grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this thesis, in whole or in part, in print, electronic, or other media, while retaining the right to use the content in future work.

Academic integrity and AI disclosure. I have disclosed all use of generative AI and other automated tools in accordance with the module policy. I remain responsible for the accuracy, originality, citations, code, data, analyses, and conclusions in this thesis.

Student name: H. M. Nipuni Tharushika Bandara

Student ID: IT22251664

Signature:  Date: 11/10/2025

Supervisor certification. The above candidate carried out this research for the bachelor's degree thesis under my supervision.

Role and name	Signature	Date
Supervisor: Ms. Karthiga Rajendran		
Co-supervisor: Ms. Thilini Jayalath		

ABSTRACT

Online recruitment fraud exposes job seekers to financial loss and identity theft, and fraudulent postings often hide behind employers that appear credible. Existing employer-verification approaches depend on a single registry or verification channel, are usually tied to the registries of other countries, and rarely explain how each signal contributed to the final decision or how the evidence weights were chosen. This research aimed to design and evaluate an explainable employer legitimacy verification component for the Sri Lankan recruitment context. The component combines three evidence layers: website analysis, multi-source registry checks (Colombo Stock Exchange, eROC/DRC, Central Bank of Sri Lanka and SLAASMB, and reputation aggregation from public platforms. A Random Forest classifier trained on 14 structured features supplies a model score, which is fused with the three-layer scores using fixed weights of 0.40, 0.30, 0.20 and 0.10 to produce a 0–100 legitimacy score, a Low, Medium or High-risk band, and a plain-language evidence trace. Weights and thresholds were examined using a domain-grouped 60/20/20 train, validation and test split of 1,514 labelled employers, with choices made on validation data and final results reported once on a held-out test set of 300 employers in an offline evaluation. The proposed weighting reached 87.0% accuracy, 90.2% F1-score and 78.1% fraud recall, outperforming equal weights (85.0% accuracy). A grid-search profile scored higher (93.0% accuracy) but was not adopted because it reduces the registry share and weakens verifiable evidence. No fraudulent employer reached the Low-risk band on the test set (0 of 107). The contribution is an explainable, registry-backed fusion design with reproducible weight and threshold evidence, supplied as a risk signal to a fraud-aware job recommender.

Keywords: employer verification, fake job detection, weighted evidence fusion, Random Forest, explainable AI

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my supervisor, **Ms. Karthiga Rajendran**, for her continuous support, insightful guidance, and encouragement throughout this research. Her expertise and constructive feedback, especially on the fusion weights and the evidence needed for the final evaluation, were instrumental in shaping the direction and quality of this work.

I am also deeply grateful to my co-supervisor, **Ms. Thilini Jayalath**, for her valuable advice and constant motivation, which helped me overcome challenges and achieve the research objectives. I am also grateful to **Mr. Koliya Pulasinghe**, the Lecturer in Charge of the Research module, for providing all the information and guidance that the project required. I sincerely thank **Mr. Saman Gunawardana**, the external validator, for reviewing this research and offering helpful industry-oriented suggestions.

I thank my group members, Kajanthan U., Sirithar Santhosh, and Chethya Methmi, for their cooperation and for helping integrate my component with the other components of the system. I declare that the design, implementation, and evaluation of the Employer Legitimacy Verification component and the preparation of this dissertation are my own work. I used an AI assistant (Claude, by Anthropic) for guidance on evaluation design, for checking analysis scripts, and for language editing, and I verified all results myself.

Heartfelt appreciation is extended to the **Department of Software Engineering, Faculty of Computing, Sri Lanka Institute of Information Technology (SLIIT)**, for providing the necessary facilities and an inspiring research environment. This research received no external funding. Finally, I wish to thank my family and friends for their unwavering support, encouragement, and understanding throughout this journey.

TABLE OF CONTENTS

1 INTRODUCTION 1
1.1 Overall project context 1
1.2 Individual component context 1
1.3 Problem statement 2
1.4 Research gap 2
1.5 Aim and objectives 3
1.4 Research questions or hypotheses 4
1.5 Scope limitations and assumptions 5
1.8 Individual contribution 5
1.9 Thesis structure 6
2 LITERATURE REVIEW AND RELATED WORK 7
2.1 Review method and scope 7
2.2 Domain and theoretical foundations 8
2.3 Related research 11
2.4 Existing systems, products and techniques 13
2.5 Gap synthesis and conceptual framework 14
3 RESEARCH METHODOLOGY 16
3.1 Research design 16
3.2 Data participants or study materials 17
3.3 Experimental or study procedure 19
3.4 Measures and evaluation metrics 20
3.5 Data analysis 22
3.6 Ethics, privacy, security and responsible AI 23
3.7 Reproducibility and research artefacts 24
4 PRODUCT AND SOLUTION DESIGN 26
4.1 Requirements 26

4.2 Component architecture within the complete solution 28
4.3 Design alternatives and decisions 30
4.4 Detailed design 32
4.5 Standards and Professional Practice 36
5 IMPLEMENTATION INTEGRATION AND TESTING 38
5.1 Development environment and technology stack 38
5.2 Component implementation 38
5.3 Integration with the other components 40
5.4 Verification and software testing 41
5.5 Deployment and operation 44
6 RESULTS AND VALIDATION 45
6.1 Research results 45
6.2 Product and technical results 51
6.3 User or stakeholder validation 52
6.4 Integration results 52
6.5 Summary of findings 53
7 DISCUSSION 54
7.1 Interpretation of findings 54
7.2 Comparison with literature baselines and existing solutions 56
7.3 Research contribution 57
7.4 Product contribution 57
7.5 Limitations and threats to validity 58
7.6 Implications 60
8 CONCLUSION AND FUTURE DIRECTIONS 61
8.1 Conclusion 61
8.2 Achievement of objectives 61
8.3 Individual research and product contributions 62

8.4 Future directions 63
REFERENCES 65
GLOSSARY 67
APPENDICES 69
Appendix A Extended results and technical specification 69
Appendix B Academic integrity and AI use disclosure 72

LIST OF FIGURES

Figure 2.1: Conceptual framework of the employer legitimacy verification component 15
Figure 3.1: Research procedure from data to final evaluation 17
Figure 4.1: Position of the employer verification component in the complete solution 29
Figure 4.2: Internal architecture of the employer verification service and its external interfaces 30
Figure 6.1: Distribution of the fused score on the test split, with the deployed bands at 45 and 70 49
Figure 6.2: Fraud recall and false positive rate against the cut-off on the validation split 49

LIST OF TABLES

Table 1.1: Research questions 4
Table 2.1: Scope of the review 8
Table 2.2: Comparison of the closest related studies with this research 12
Table 2.3: Existing techniques for checking an employer 13
Table 2.4: Gaps identified in the literature and how this research addresses them 14
Table 3.1: Research questions and the methods used to answer them 16
Table 3.2: Structured features used by the model 18
Table 3.3: Evaluation measures 20
Table 3.4: Objectives, metrics and evidence 21
Table 3.5: Research artefacts 24
Table 4.1: Functional requirements 26
Table 4.2: Non-functional requirements 27
Table 4.3: Interfaces of the component 29
Table 4.4: Design decisions 30
Table 4.5: Lists the tools and technologies that were used and why 32
Table 4.6: Main fields of the /predict response 32
Table 4.7: Scoring rules of the deployed layer 34
Table 4.8: Code structure of the service 36
Table 5.1: Development environment 38
Table 5.2: Main settings of the service (environment variables and defaults) 39
Table 5.3: Automated tests and their results 41
Table 5.4: Scenario tests of the service and their results 42
Table 6.1: Random Forest results by split (cut-off 0.5 on the model score) 45
Table 6.2: Weight profiles on the validation and test splits (cut-off 45) 46
Table 6.3: Test results with Wilson 95% intervals (n = 300) 46
Table 6.4: AUC on the test split by evidence source (primary data) 47
Table 6.5: Threshold sweep on the validation split (selected rows) 48
Table 6.6: Risk bands on the test split (number of legitimate/fraudulent employers) 48
Table 6.7: Mean score by class on the test split 50
Table 6.8: Test results on the as-collected feature table (cut-off 45, n = 300) 50
Table 6.9: Outcome of the hypotheses 51
Table 6.10: Technical results of the service 52

Table 6.11: Summary of findings 53
Table 7.1: Comparison with the closest studies 56
Table 8.1: Achievement of the objectives of Section 1.5.2 62
Table A.1: Settings of the model, the fusion and the bands 69
Table A.2: Fraud recall of the weight profiles on the test split 69
Table A.3: AUC on the test split by source and by combination 70
Table A.4: Risk bands on the test split (genuine/fraudulent employers) 70
Table B.1: Use of AI tools 72

LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
AUC	Area Under the ROC Curve
BOI	Board of Investment of Sri Lanka
CBSL	Central Bank of Sri Lanka
CSE	Colombo Stock Exchange
DRC	Department of Registrar of Companies
eROC	Electronic Registrar of Companies portal
FPR	False Positive Rate
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
IRCSL	Insurance Regulatory Commission of Sri Lanka
JSON	JavaScript Object Notation
ML	Machine Learning
NFR	Non-Functional Requirement
FR	Functional Requirement
REST	Representational State Transfer
RF	Random Forest
RQ	Research Question
SLAASMB	Source of the list of Specified Business Enterprises (slaasmb.gov.lk)

Abbreviation	Meaning
SME	Small and Medium Enterprise
SLIIT	Sri Lanka Institute of Information Technology
URL	Uniform Resource Locator

1 INTRODUCTION

1.1 Overall project context

Online recruitment fraud reaches a job seeker at several points: the job advertisement, the employer behind it, the conversation with the recruiter, and the choice of which jobs to apply for. Most existing tools protect only one of these points, usually by classifying the text of an advertisement. The group project R26-SE-002 therefore proposes an AI ecosystem of four connected components, each producing an independent risk signal.

- Fake job post detection: a multilingual XLM-RoBERTa model classifies English and Sinhala advertisements and gives a three-tier suspicion score explained with SHAP and LIME.
- Employer legitimacy verification: the component reported in this dissertation.
- Scam communication detection: the Behaviour-Inspired Semantic Tactic Detection (BSTD) framework uses a fine-tuned DistilBERT model to identify manipulation tactics in recruiter-candidate conversations.
- Fraud-aware job recommendation: TF-IDF skill matching is combined with the risk signals of the other components, and jobs are ranked with the Entropy Weight Method and TOPSIS.

The outputs of the three detection components feed the recommendation component. Each group member designed, implemented and evaluated one component.

1.2 Individual component context

The employer verification component answers a question that text analysis cannot answer reliably: does the organization named in the advertisement really exist, and does the evidence about it look like that of a genuine employer? It collects evidence in three layers. Website analysis checks HTTPS, organizational pages and payment-risk or urgency language. Registry checks look the company up in Sri Lankan sources: the Colombo Stock Exchange (CSE), eROC/DRC company records, the Central Bank of Sri Lanka (CBSL) and the SLAASMB list of Specified Business Enterprises.

Reputation aggregation uses public platforms such as LinkedIn, Glassdoor, Indeed and Trustpilot, and reports of scams lower the score.

A Random Forest trained on 14 structured features gives a model score, which is combined with the three-layer scores by weighted fusion (weights 0.40, 0.30, 0.20 and 0.10). The result is a score from 0 to 100, a risk band (Low, Medium or High), and an evidence trace in plain language, which is passed to the recommendation component. A missing registry entry is treated as "not confirmed" and not as proof of fraud, because many genuine small and medium enterprises (SMEs) are not listed in public registers.

1.3 Problem statement

Job seekers have no easy and reliable way to check the legitimacy of an employer before sharing personal data or paying money. Manual checking requires knowing which registers exist and how to judge websites and reviews, and fraudsters exploit this by copying real company names, building professional-looking sites, and using urgency and payment requests. Automated approaches leave open the following problem for the Sri Lankan setting: there is a need for an explainable employer-verification method that combines website, registry and reputation evidence, that does not wrongly penalize genuine SMEs for missing registry entries, and whose evidence weights and risk thresholds are justified by reproducible evaluation instead of assumption.

1.4 Research gap

Mahbub et al. [1] showed that registry and website features raised Random Forest accuracy from 89.22% to 91.86%. Alghamdi and Alharby [2] found employer metadata to be among the strongest indicators of legitimacy. Pangare et al. [3] showed that hybrid verification with identity checks and reputation data outperforms text-only classification. Taken together, these studies show that employer-level evidence is valuable, but they leave four gaps:

1. Local registry coverage: existing work is tied to registries of other countries, and none combines the CSE, eROC/DRC, CBSL and SLAASMB sources.

2. Single-channel verification: many approaches rely on one channel, with no weighted combination of independent evidence.
3. Explainability: systems do not show how much each signal contributed to the decision.
4. Justification of weights and thresholds: these are rarely tested on separate validation and test data, and the accuracy lost by prioritizing explainability is rarely reported.

1.5 Aim and objectives

1.5.1 Aim

The aim is to design, implement and evaluate an explainable, multi-source employer legitimacy verification component for the Sri Lankan recruitment context, with weights and risk thresholds justified by held-out evidence.

1.5.2 Objectives

The objectives are:

O1. Review related work on employer verification and evidence fusion and identify the gaps (Section 1.4 and Chapter 2).

O2. Design a multi-source evidence pipeline of website, Sri Lankan registry and reputation layers.

O3. Build a Random Forest model on structured employer features and measure it on held-out data.

O4. Define a weighted fusion score and determine its weights by a documented, reproducible experiment.

O5. Define Low, Medium and High-risk bands and test whether they keep fraudulent employers out of the Low band.

O6. Expose the component as a service that returns the score, band and evidence report, and integrate it with the recommendation component.

1.4 Research questions or hypotheses

The research is guided by the four questions listed in Table 1.1.

1.6.1 Research questions

Table 1.1: Research questions

No.	Research question
RQ1	Which website, registry and reputation signals can be collected automatically for Sri Lankan employers, and how can they be turned into reliable features?
RQ2	How well can a Random Forest model on these features separate legitimate from fraudulent employers on data it has not seen?
RQ3	How should the weights of the fusion score be determined, and how much accuracy is lost when explainability and registry evidence are prioritized?
RQ4	Do the Low, Medium and High-risk bands keep fraudulent employers out of the Low band, and what do the bands cost in false alarms?

1.6.2 Hypothesis

To make the evaluation testable, the following working hypotheses were set before the experiments were run.

- H1. The proposed weights (0.40/0.30/0.20/0.10) give better held-out accuracy and fraud recall than equal weights (0.25 for each signal).
- H2. No single evidence source is sufficient: profiles that rely mainly on the registry, or that ignore reputation, perform clearly worse on fraud recall than the balanced profile.
- H3. With the conservative thresholds of 45 and 70, few or no fraudulent employers on the held-out test set are placed in the Low-risk band.

The results chapter reports whether each hypothesis was supported. A weight profile chosen only by accuracy is expected to score higher than the proposed profile; the research question RQ3 asks how large that difference is, so that the decision to keep the proposed weights can be defended with evidence.

1.5 Scope limitations and assumptions

1.7.1 Scope

The component verifies employers that operate in or recruit from Sri Lanka, using public sources only, and returns a risk estimate and not a legal ruling. The evaluation covers the model score, the weighted fusion and the risk bands; the other components are outside the scope.

1.7.2 Limitations

The evaluation uses one held-out test split of 300 employers, so confidence intervals are wide. The dataset was constructed for the project with controlled randomization of some feature values, so results describe controlled offline performance. Registry portals may change or limit automated access. The deployed scoring service uses a capped point system and is not identical to the normalized formula used in the evaluation. The model score is not a calibrated probability.

1.7.3 Assumptions

A genuine employer leaves some verifiable public trace; a missing registry match means “not confirmed”; and the labels in the dataset are correct.

1.8 Individual contribution

- Designed the three-layer evidence pipeline and implemented the registry checks for CSE, eROC/DRC, CBSL and SLAASMB support and merging of results from several sources.
- Defined the 14 structured features, prepared the labelled dataset (1,514 employers) and trained the Random Forest model.
- Designed the weighted fusion score, the risk bands and the plain-language evidence trace.
- Designed and ran the evaluation of weights and thresholds on a domain-grouped train/validation/test split, with a single final test, cost-sensitive threshold analysis and confidence intervals, using reproducible scripts.
- Exposed the component as a service for integration with the recommendation component.

Link to the group solution. The component takes the employer's name and website from a job advertisement and returns the score, risk band and evidence trace, which the recommendation component uses as one of the risk signals when it ranks jobs. The main findings are reported in Chapter 6.

1.9 Thesis structure

Chapter 2 reviews the literature and ends with a conceptual framework. Chapter 3 describes the research methodology. Chapter 4 presents the product and solution design, and Chapter 5 the implementation, integration and testing. Chapter 6 reports the results and validation, Chapter 7 discusses them, and Chapter 8 gives the conclusion and future directions. The dissertation ends with the references, a glossary and the appendices.

2 LITERATURE REVIEW AND RELATED WORK

This chapter reviews the research and the existing practice that bear on employer legitimacy verification. It explains how the review was carried out, sets out the domain and theoretical foundations that the component relies on, examines related research and existing systems, and ends by synthesizing the gaps into a conceptual framework that guides the methodology in Chapter 3.

2.1 Review method and scope

The review looked for work that helps answer the research questions in Section 1.6: which employer-level signals can separate legitimate from fraudulent employers, how several signals are combined, how scores are turned into decisions, and how such systems are evaluated. Three strands were reviewed: (a) online recruitment fraud and fake job detection, (b) employer-level evidence such as registries, websites and reputation, and (c) methods for combining evidence and choosing thresholds, including classifier combination, cost-sensitive decision making and confidence intervals.

Searches were conducted in Google Scholar and IEEE Xplore using the terms "fake job detection", "employer verification", and "online recruitment fraud". Studies were included when they addressed recruitment fraud detection, employer authenticity verification, or a method directly applied in this dissertation, including classifier combination, cost-sensitive decision making, probability calibration, and confidence interval estimation. Studies that only classified job advertisement text were included as background context rather than direct comparators. The search identified eleven works selected for their direct relevance to the research questions: studies on contextual employer verification and registry-based fraud detection [1] [2], multi-layered employer authenticity systems [3], the Random Forest ensemble method used as the primary classifier [4], probability calibration for supervised classifiers [5], classifier combination theory underlying the hybrid fusion approach [7][8], model explainability methods applied to the evidence trace [9][10], and statistical methods for evaluating classification performance under uncertainty [11]. No published study

was found that verifies Sri Lankan employers against Sri Lankan registries specifically; this is stated as a finding of this review and not as proof that none exists.

Table 2.1: Scope of the review

Strand	Question asked	Used in
Recruitment fraud and fake job detection	What forms does the fraud take and what has been automated?	Sections 2.2.1 and 2.3.1
Employer-level evidence	Which registry, website and reputation signals separate real from fake employers?	Sections 2.2.2, 2.3.2 and 2.4
Evidence combination and decision making	How are signals combined, thresholds set and results evaluated?	Sections 2.2.3 to 2.2.6

2.2 Domain and theoretical foundations

2.2.1 Online recruitment fraud

Online recruitment fraud (ORF) is the practice of using fake job advertisements, fake employers, or fake recruiters to obtain money, personal data, or labor from job seekers. Common forms are advance-fee scams, in which the applicant pays for registration, training or travel; identity harvesting, in which documents and bank details are collected; and impersonation of genuine companies. Because the fraudster controls the advertisement and often the website, any signal that originates only from the advertiser can be copied. Signals that originate from third parties, such as official registers and independent review platforms, are harder to forge and are therefore central to employer verification.

2.2.2 Employer legitimacy and its evidence sources

Employer legitimacy can be judged from three kinds of evidence. Registry evidence shows whether the organization is recorded by an authority. In Sri Lanka, the relevant sources include company registration records held by the Department of Registrar of Companies (eROC/DRC), the Colombo Stock Exchange (CSE) for listed companies, the Central Bank of Sri Lanka (CBSL) for licensed financial institutions, and the list of Specified Business Enterprises maintained for financial-reporting oversight (SLAASMB). Reputation evidence shows whether independent parties know the organization, for example through employee and customer reviews on platforms such

as Glassdoor, Indeed and Trustpilot, professional network profiles such as LinkedIn, Sri Lankan job and news sources such as TopJobs.lk and the Daily FT, and reports of scams. Website evidence describes how the organization presents itself: use of HTTPS, presence of About, Contact, Privacy and Terms pages, and the language used on the page.

Each source has limits. A registry entry proves existence but not honesty, and genuine small firms may be missing from some registers. Reputation signals may be absent for new firms and can be manipulated. Website features are cheap to produce. A sound verification method therefore combines the sources and treats a missing entry as "not confirmed" and not as proof of fraud.

2.2.3 Random Forest classification

A Random Forest builds many decision trees on random samples of the data and random subsets of the features, and combines their votes, and its main settings are the number and the depth of the trees [4]. It handles mixed binary and numeric features, captures non-linear relationships, resists overfitting, and provides feature importance values, making it a common choice for structured fraud-detection data. In this research, it is used on 14 structured employer features. The average class probability across the trees is a useful score, but it is not automatically a calibrated probability; whether the scores of a Random Forest can be read as probabilities depends on how the forest is tuned, and recent work compares calibration methods for Random Forests in detail [5]. Because this was not verified for the present data, the output is called a model score. For this reason, we call the model output a model score unless we apply calibration.

2.2.4 Combining evidence

Combining several sources of evidence is a long-standing topic in pattern recognition. Surveys of ensemble learning show that simple fixed rules for combining classifiers, such as averaging, can be robust when individual estimates are noisy [6]. A weighted sum is a direct extension of the averaging rule in which more reliable sources receive more weight. Its advantage for a user-facing system is transparency: each weight can be stated and each source's contribution can be shown. Its weakness is that the weights must be justified. If they are set by assumption, the result may only look precise; if they are tuned only to maximize a metric on one dataset, they may favor whichever

signal is most informative in that dataset and remove evidence that users can check for themselves.

2.2.5 Thresholds, costs and risk bands

A score becomes a decision only when thresholds are applied. The usual default of treating 0.5 as the boundary assumes that both kinds of error cost the same. In cost-sensitive learning, when the cost of a missed positive differs from the cost of a false alarm, the best decision threshold follows from the ratio of those costs and not from the class balance [7]. Reviews of learning from imbalanced data argue that performance should be judged with measures that suit unequal classes and costs, and not by accuracy alone [8]. This is the reason for presenting recall and false-alarm rates over a range of thresholds, as is done later in this dissertation, and for using three risk bands, which allow uncertain cases to be sent for manual checking instead of being forced into “safe” or “fraud”.

2.2.6 Explainability

Explainability methods such as SHAP [9] and LIME [10] describe why a model made a prediction by attributing the output to input features. They are valuable for text models whose internal reasoning is opaque. For a score built from named evidence sources the explanation can be more direct: the system can list the sources that were consulted, what each returned, and how much each contributed to the final score. This favours a fusion design with fixed, visible weights, evidence reports and a score breakdown written in plain language, and it makes the justification of those weights part of the research. The component in this dissertation takes this approach and does not use SHAP or LIME; it also inspects Random Forest feature importance during model development to confirm that the model relies on meaningful signals.

2.2.7 Evaluation methodology

Reliable evaluation needs data that the model has not seen during training or tuning. When several records come from the same source, a random split can place near-duplicates in both training and test sets and overstate performance, so splits should be grouped by source (here, by website domain). Tuning choices such as weights and

thresholds should be made on a validation set, and the test set should be used once. Because test sets for fraud are small, a performance figure should be accompanied by a confidence interval; the Wilson score interval is recommended over the simple normal approximation for proportions, especially when counts are small or proportions are near 0 or 1 [11]. The EMSCAD collection of Vidros et al. [17] is the usual public benchmark for such work.

2.3 Related research

2.3.1 Fake job posting detection

Most published work on recruitment fraud classifies the text of a job advertisement as real or fake, usually using public benchmark collections of postings and classical or deep learning classifiers. This line of research shows that language and posting-level features carry a strong signal. Its limitation for the present problem is that the classifier judges the advertisement and not the organization behind it, so a well-written advertisement from a fake employer can pass. These approaches are complementary to employer verification and correspond to the first and third components of the group project.

2.3.2 Employer-level and registry-based verification

Three studies are closest to this research. Mahbub et al. [1] proposed a multi-layer fraud detection framework that checks employer authenticity against both government and non-government registries. Their results showed that adding contextual features, such as whether a company website exists and whether the company appears in a registry, raised Random Forest accuracy from 89.22% to 91.86%. This supports the use of registry and website evidence as features. Alghamdi and Alharby [2] found that employer metadata, including the presence of a company profile, a corporate logo and an industry classification, was among the strongest indicators separating legitimate from fraudulent employers, which supports the use of structured employer-level signals. Pangare et al. [3] described JobScamShield, a system that combines GST number validation, CEO identity checks, Glassdoor reputation data and NLP of job descriptions, and reported that this kind of hybrid verification outperforms text-only classification.

These studies agree that employer-level evidence improves detection. They differ from the present work in four respects, summarized in Table 2.2: they are built around registries or identifiers of other countries, they rely on a limited set of verification channels, they do not show how much each signal contributed to the decision, and they do not report how combination weights or decision thresholds were chosen and tested.

Table 2.2: Comparison of the closest related studies with this research

Study	Evidence used	Reported finding	Limitation relative to this research
Mahbub et al. [1]	Government and non-government registries; website existence; contextual features	Random Forest accuracy rose from 89.22% to 91.86% with contextual features	Not designed for Sri Lankan registries; no weighted fusion or threshold analysis reported
Alghamdi and Alharby [2]	Employer metadata: company profile, logo, industry	Employer metadata among the strongest indicators	Metadata only; no registry or reputation checks
Pangare et al. [3]	GST validation, CEO identity checks, Glassdoor data, NLP of descriptions	Hybrid verification outperformed text-only classification	Tied to Indian identifiers; weights and thresholds not justified
This research	Sri Lankan registries, reputation platforms, website features, Random Forest score	Reported in Chapter 6	Weights and thresholds tested on held-out data; explainable evidence trace

2.3.3 Evidence fusion and decision making in related fields

Weighted combination of evidence is used widely in credit scoring, spam filtering and risk assessment, where several indicators are merged into one score and mapped to risk categories. The general lesson from the classifier-combination literature [6] is that combined estimates are usually better than single ones, and from the cost-sensitive literature [7], [8] that thresholds should reflect the cost of errors. What is rarely reported in recruitment-fraud studies is the step between the two: how the weights of a fusion score were chosen, whether they were tested on data separate from the data used to choose them, and what accuracy is traded away when interpretability is given priority. This dissertation addresses that step explicitly.

2.3.4 The Sri Lankan context

Sri Lankan job seekers face the same fraud patterns as elsewhere, including fake offers circulated on social media and pyramid-style schemes presented as employment. Official sources that can be used to check whether an employer exists are described in Section 2.2.2. The review did not find a published employer-verification method that combines these local sources, and the Sri Lankan work found on fraudulent content concerned language-processing problems, such as detecting fake news in Sinhala, and not employer verification. This is a finding of the search described in Section 2.1 and not a claim that no such work exists.

2.4 Existing systems, products and techniques

Beyond research papers, employers can be checked using tools and practices that already exist. These are described in Table 2.3 together with their strengths and weaknesses for a job seeker.

Table 2.3: Existing techniques for checking an employer

Technique	What it checks	Strength	Weakness
Manual registry search (eROC, CSE, CBSL, SLAASMB lists)	Whether the company is recorded by an authority	Authoritative evidence of existence	Needs knowledge of which register to use; separate portals; some require manual lookup; not every genuine firm is listed
Job-portal employer verification badges	Whether the portal has checked the advertiser	Convenient for users of that portal	Applies only to that portal; criteria are usually not published; does not cover social-media advertisements
Review and reputation sites (for example Glassdoor, Trustpilot)	Employee and customer opinions	Independent third-party view	Missing for new or small firms; reviews can be manipulated
Web search and scam-report lookup	Public warnings about the company or domain	Can reveal known scams	Relies on the user knowing what to search; new scams are not yet reported
Domain and website inspection (HTTPS, domain age, page content)	Technical and presentational signals	Cheap and automatic	Easy for fraudsters to imitate

Text-based fake job classifiers	Language of the advertisement	Fast and scalable	Judges the advertisement, not the organization
Proposed component	Registry, reputation, website and model score fused with fixed weights	Combines sources, returns an evidence report and score breakdown, treats missing entries as "not confirmed"	Depends on the availability of public data and on registry portals remaining accessible

Each existing technique covers part of the problem. A job seeker who wants a reliable answer would have to use several of them and judge the combined result alone. The proposed component automates that combination and returns the reasons together with the score.

2.5 Gap synthesis and conceptual framework

2.5.1 Synthesis of gaps

Table 2.4 links each gap found in the review to the part of this research that addresses it.

Table 2.4: Gaps identified in the literature and how this research addresses them

Gap	Evidence from the review	How this research addresses it
G1 No Sri Lankan registry coverage	Related work uses registries or identifiers of other countries (Section 2.3.2); no Sri Lankan study found (Section 2.3.4)	Registry layer built on eROC/DRC, CSE, CBSL and SLAASMB, with BOI and IRCSL as supplementary sources
G2 Reliance on few channels	Single-channel or limited-channel verification (Section 2.3.2)	Three evidence layers plus a model score combined by weighted fusion
G3 Unexplained contributions	Existing systems do not show how much each signal contributed (Sections 2.3.2 and 2.4)	Fixed visible weights, evidence reports listing sources and findings, a score breakdown per source, and plain-language reasons
G4 Unjustified weights and thresholds	Weights and thresholds rarely tested on separate data (Section 2.3.3)	Domain-grouped train/validation/test split, five weight profiles and a grid search, threshold sweep with cost analysis, single final test

G5 Treatment of missing evidence	Registry absence can penalize genuine small firms (Section 2.2.2)	Absence treated as “not confirmed”; three risk bands with a medium band for manual verification

2.5.2 Conceptual framework

The conceptual framework in Figure 1 combines the foundations of Section 2.2 with the gaps in Table 2.4. The input is a company name and, where available, a website address. Three evidence layers (website, registry and reputation) collect signals. The signals are converted into 14 structured features that train a Random Forest, which supplies the model score. The model score and the three-layer scores are combined by weighted fusion with weights of 0.40, 0.30, 0.20 and 0.10 to give a score from 0 to 100, which is mapped to a Low, Medium or High-risk band and returned with an evidence report, a score breakdown and plain-language reasons. The band and score are passed to the fraud-aware job recommendation component. The weights and thresholds are tested on held-out data, which is shown in the figure as a dashed link, because the review showed this step to be the weakest in earlier work.

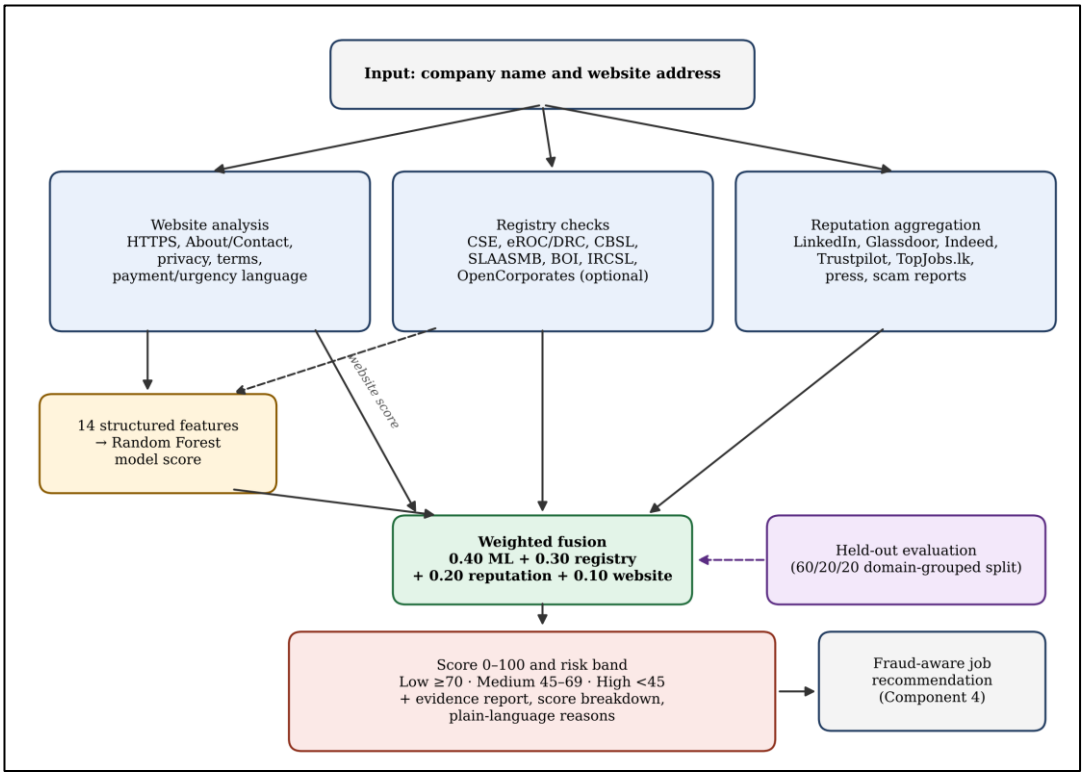


Figure 2.1: Conceptual framework of the employer legitimacy verification component

3 RESEARCH METHODOLOGY

This chapter explains how the research was designed and carried out: the research design, the data, the experimental procedure, the measures, the analysis, the ethical and security considerations, and the artefacts that make the work reproducible. The design of the product itself is described in Chapter 4 and its implementation in Chapter 5.

3.1 Research design

The research follows a quantitative design-and-evaluate approach. A solution (the employer legitimacy verification component) was designed from the foundations in Chapter 2, and its central design decisions, namely the model, the fusion weights and the risk thresholds, were evaluated experimentally on labelled employer data. The evaluation is a controlled offline experiment on a dataset of labelled employers. No human participants were involved in the experiment.

The design addresses the research questions of Section 1.6 as shown in Table 3.1. Choices that could affect the results, such as the weights and thresholds, were made using one part of the data (the validation split) and judged on another (the test split), so that the reported performance is not an artefact of tuning.

Table 3.1: Research questions and the methods used to answer them

Research question	Method used	Section
RQ1: Which signals can be collected and turned into features?	Design of the website, registry and reputation layers; definition of 14 structured features	3.2 and Chapter 4
RQ2: How well does the Random Forest separate legitimate from fraudulent employers on unseen data?	Domain-grouped 60/20/20 split; model trained on the training split only; metrics on the test split	3.3 and 3.4
RQ3: How should the fusion weights be determined, and what is the cost of prioritizing explainability?	Five weight profiles and a grid search compared on validation data; final comparison on test data	3.3 and 3.5
RQ4 Do the risk bands keep fraud out of the Low band, and at what cost in false alarms?	Threshold sweep with cost-sensitive analysis on validation data; risk-band table and Wilson confidence intervals on test data	3.3 to 3.5

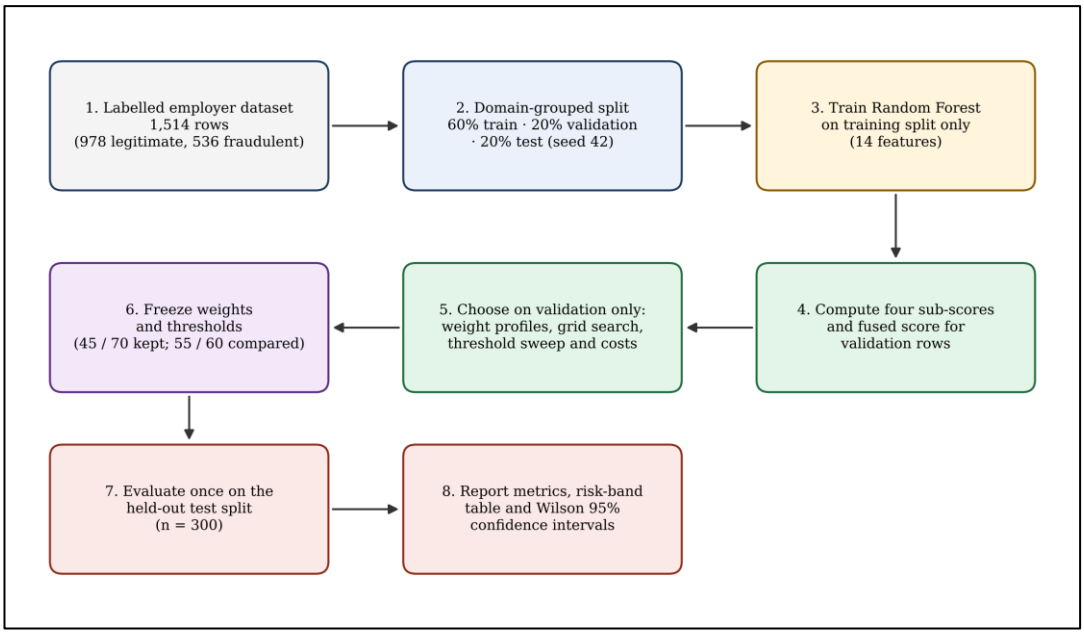


Figure 3.1: Research procedure from data to final evaluation

3.2 Data participants or study materials

3.2.1 Participants

The experiment used no human participants and no personal data about individuals. Each record describes an organization (a company name and its website) and a label. Validation of the product by stakeholders is reported separately in Chapter 6.

3.2.2 Dataset

The dataset contains 1,514 labelled employer records: 978 legitimate and 536 fraudulent. Each record holds the structured features extracted by the website, registry and reputation layers, and a binary label. In the dataset files the label 1 means legitimate and 0 means fraudulent, so the model output is the score for the class “legitimate”.

The dataset is partly generated, and this is stated openly because it limits what the results mean. Legitimate records were built from company evidence collected from public sources. In the final construction step, the feature values of the fraudulent records, and the payment-risk and urgency-language flags of the legitimate records, were redrawn at fixed rates chosen by the author so that the fraudulent class resembles active scam websites with controlled overlap with legitimate ones. An earlier version

of the data contained artefacts, for example features that separated the classes only because of how the data were gathered (such as the age of the domain, which was constant for the legitimate class); these were removed from the model inputs. The results are therefore a controlled evaluation of the method and not a measurement of accuracy on real-world scam traffic. Fraudulent organizations were identified from two public sources: Sri Lankan news reports about pyramid-type companies that were banned or acted against, and comments on social-media posts in which job seekers reported being defrauded by a named company. Legitimate organizations and the website evidence for both classes were compiled manually from public websites. Labels that rest on news reports are stronger than labels that rest on social-media comments, which are unverified reports and may be mistaken or malicious; this is a limitation of the labels and is discussed in Chapter 7. Only company-level information was recorded, and no personal data of victims was collected.

3.2.3 Features

The Random Forest uses the 14 features in Table 3.2. They describe the website, the registry result and the reputation signals.

Table 3.2: Structured features used by the model

Group	Feature	Meaning
Website	has_https	The site is served over HTTPS
Website	is_http_only	The site is served only over HTTP
Website	subdomain_count	Number of subdomain levels in the address
Website	has_about, has_contact	An About page and a Contact page exist
Website	has_privacy_policy, has_terms	A privacy policy and terms of use exist
Website	has_payment_risk	The content asks for fees or payments in a way typical of scams
Website	has_urgency_language	The content uses pressure or urgency wording
Website	content_score	Overall content quality score of the site
Registry	is_registered	The company was found in at least one official source

Reputation	has_glassdoor, has_indeed, has_trustpilot	Presence of the company on each review platform

3.2.4 Data splits

The data were divided into a training split (60%), a validation split (20%) and a test split (20%). The split is grouped by website domain, so that all records that share a domain fall in the same split and near-duplicate websites cannot appear in both training and testing. The random seed was fixed at 42. The training split has 901 employers (578 legitimate and 323 fraudulent), the validation split has 313 (205 legitimate and 108 fraudulent), and the test split has 300 (195 legitimate and 105 fraudulent). Because the split is grouped by domain, the proportions are close to, but not exactly, 60/20/20.

3.3 Experimental or study procedure

The procedure has eight steps, shown in Figure 2.

1. Prepare the dataset and remove inputs that would let the model learn artefacts of data collection.
2. Split the data 60/20/20 by domain with seed 42.
3. Train a Random Forest classifier on the training split only. The settings are 200 trees, a maximum depth of 8, a minimum of 16 samples to split a node, a minimum of 8 samples per leaf, and balanced class weights [4]. The output for each employer is the average probability of the class “legitimate” across the trees, called the model score P_ML.
4. Compute the three-layer scores for every record: the registry score from the registration result, the reputation score from the review-platform signals, and the website score from the HTTPS and organizational-page signals. Each lies between 0 and 1.
5. Combine the four scores with the fusion formula: Score = 100 × (w1 × P_ML + w2 × R_registry + w3 × R_reputation + w4 × R_website), where w1 to w4 sum to 1. Five weight profiles were compared: the proposed baseline (0.40, 0.30, 0.20, 0.10), ML-dominant (0.70, 0.10, 0.10, 0.10), registry-heavy (0.20, 0.60, 0.10, 0.10), equal (0.25 each), and reputation-ignored (0.50, 0.40, 0, 0.10). A grid

search over all weight combinations in steps of 0.05 was run on the validation split.

6. Sweep thresholds from 20 to 80 in steps of 5 on the validation split, and record fraud recall, false positive rate, precision, and the weighted cost of errors for cost ratios of 3:1, 5:1, and 10:1 (a missed fraud costing 3, 5, or 10 times a false alarm) [7].
7. Freeze the weights and thresholds. The proposed weights (0.40, 0.30, 0.20, 0.10) and thresholds (45 and 70) are kept as the deployed design; the best grid-search weights and the validation-selected thresholds (55 and 60) are retained only as comparisons.
8. Evaluate the frozen configurations once on the test split and report metrics, the risk-band table and confidence intervals.

The Medium and Low boundaries of the data-driven comparison were selected by two rules on the validation split. The boundary between High and Medium is the lowest threshold at which at least 90% of fraudulent employers score below it. The boundary between Medium and Low is the lowest threshold above which at least 90% of employers scoring at or above it are legitimate. The proposed thresholds of 45 and 70 were set before the experiments, as conservative boundaries intended to bias against calling a fraudulent employer safe, and the experiment checks them; it does not claim that they were optimized.

3.4 Measures and evaluation metrics

Legitimate is treated as the positive class for classification metrics, and fraud is treated as the class of interest for risk metrics. Table 3.3 defines the measures.

Table 3.3: Evaluation measures

Measure	Definition	Why it is used
Accuracy	Correct decisions divided by all decisions	Overall agreement with the labels
Precision	Of employers judged legitimate, the share that are legitimate	Cost of wrongly approving an employer
Recall	Of legitimate employers, the share judged legitimate	Avoiding needless rejection of genuine employers

F1-score	Harmonic mean of precision and recall	Single summary of the two
Fraud recall	Fraudulent employers scored below the cut-off divided by all fraudulent employers	Direct measure of how much fraud is caught
False positive rate (FPR)	Legitimate employers wrongly flagged divided by all legitimate employers	Burden on genuine employers
Fraud acceptance rate	Fraudulent employers scored at or above the cut-off divided by all fraudulent employers (equals 1 minus fraud recall)	Risk of approving fraud
Legitimate precision at or above t	Share of employers scoring at least t that are legitimate	Used to set the Medium/Low boundary
Weighted error cost	C_FN × missed frauds + C_FP × false alarms, with C_FN: C_FP of 3:1, 5:1 and 10:1	Chooses thresholds when errors cost different amounts [7], [8]
Wilson 95% interval	Confidence interval for a proportion [11]	Shows uncertainty on a small test set

The term “false positive rate” is used here only for legitimate employers wrongly flagged. In an earlier version of the analysis, the same words described the share of fraud that was approved; that figure is now named the fraud acceptance rate, and all reported results use the definitions in Table 3.3.

Every objective of Section 1.5 is linked to a metric and to the evidence that will show whether it was met, as in Table 3.4. An objective without evidence cannot be defended, so the table also shows where the evidence is reported

Table 3.4: Objectives, metrics and evidence

Objective	Research question	Metric	Evidence	Reported in
O1 Review related work and identify gaps	Motivates all	Gaps G1 to G5	Literature review and comparison tables	Chapter 2
O2 Multi-source evidence pipeline	RQ1	Number of features and layers implemented; sources covered	Feature table, architecture and component tests	Chapters 4 and 5
O3 Random Forest on employer features	RQ2	Accuracy, precision, recall, F1 on the test split	Test-split result against the validation result	Chapter 6

O4 Fusion weights found by experiment	RQ3	Fraud recall, FPR and weighted cost per profile	Five profiles and grid search compared on validation, then on test	Chapter 6
O5 Risk bands that keep fraud out of Low	RQ4	Fraud in the Low band; fraud recall; legitimate precision; Wilson 95% interval	Risk-band table on the test split	Chapter 6
O6 Service and integration	Supports RQ1 to RQ4	Pass rate of functional tests; response time of the service; correct hand-over to the recommendation component	Test results and an integration test	Chapter 5, validation in Chapter 6

3.5 Data analysis

The analysis compares the weight profiles on validation data, applies the selection rules for the risk-band boundaries, and reports the final comparison on test data. For each profile, the metrics of Table 3.4 are calculated from the confusion matrix at the decision cut-off. Because the main decision is whether an employer is flagged, flagged means a fused score below the Medium/Low boundary.

Three working hypotheses were set before the experiments: H1, the proposed weights outperform equal weights on held-out data; H2, profiles that rely mainly on registry evidence or ignore reputation perform clearly worse on fraud recall; and H3, with thresholds of 45 and 70, few or no fraudulent employers reach the Low band. A hypothesis is treated as supported when the held-out result points in the stated direction and the difference is larger than the width of the confidence interval where an interval is reported; with a single test split, no further significance test is claimed.

Confidence intervals use the Wilson score interval. For a proportion p observed on n cases and z = 1.96, the interval is centered at $(p + z^2/2n) / (1 + z^2/n)$ with half-width $z \times \sqrt{(p(1-p)/n + z^2/4n^2)} / (1 + z^2/n)$ [11]. Cost-sensitive analysis counts missed frauds and false alarms at each threshold for the three cost ratios and records the threshold with the lowest total cost.

Baselines. A result is only meaningful against a reference. The equal-weight profile (0.25 each) is the baseline that assumes no knowledge of which signal is more useful, the ML-dominant profile represents the model-centered approach of most related work, and the registry-heavy and reputation-ignored profiles test the importance of single sources. The proposed profile is therefore not reported as simply better or worse. For each comparison, the results show where it improves, where it does not, and whether the trade-off is acceptable, in particular the accuracy given up for an explainable fixed-weight design.

From data to findings. The raw outputs are the confusion counts for each profile and threshold. They are summarized as the metrics of Table 3.4 (the results), interpreted against the hypotheses and the research questions (the findings), and used in Chapter 8 to state what the work adds (the contribution). Chapter 6 reports what happened, and Chapter 7 explains why and what it means.

Validity. Four threats are considered. Measurement: the labels come from news reports, social-media reports and manual collection, so a label may be wrong, and fraud recall is therefore a measure against these labels and not against all real fraud. Fair comparison: all profiles use the same model, the same split and the same test set, and tuning is done only on the validation split. Generalization: the dataset is limited to 1,514 employers, part of it was constructed for the project, and the test split has 300 employers, so the results may not carry over to other settings. Repeatability: the seed, the split and the scripts are fixed (Section 3.7). These threats are stated in the limitations of Section 1.7 and discussed in Chapter 7.

3.6 Ethics, privacy, security and responsible AI

3.6.1 Ethics and privacy

The study did not collect data from human participants. All evidence comes from public sources: public registers, public review platforms and public websites. The data describe organizations, not private individuals, and the dataset contains no personal contact details. No paid or restricted databases were accessed.

3.6.2 Security

Security. Access keys for external services are kept in environment settings and are not written into source code or shared files. The component only reads public sources and does not log user credentials. Requests to external sites are made through the service and not by the user's browser.

3.6.3 Responsible AI

A risk result about a named company can harm a genuine business if it is wrong. The component therefore presents its output as a risk estimate with reasons, not as an accusation. A missing registry match is treated as "not confirmed" and the medium band asks the user to verify, for example, on the eROC portal. The weights are fixed and visible so users can explain and challenge the results. The model score is a classifier output and is not presented as a calibrated probability. The limitations of the data, including the generated part, are reported in Section 3.2 and discussed in Chapter 7. The use of AI tools in preparing this dissertation is disclosed in Appendix B.

3.7 Reproducibility and research artefacts

The experiments can be repeated from the following artefacts, all of which use the fixed seed of 42 and the same split.

Table 3.5: Research artefacts

Artefact	Purpose
data/final_realistic_dataset.csv	Labelled employer dataset (1,514 rows)
models/final_realistic_model.pkl	Trained Random Forest model used by the service
app/weight_sensitivity_analysis.py	Compares the five weight profiles and the grid search; writes the sensitivity report
notebooks/threshold_analysis.py	Domain-grouped split, retraining on the training split, threshold sweep, cost analysis, risk-band table and plots
outputs/threshold/*.csv and *.png	Validation sweep, weight-profile comparison on validation and test, band distribution, threshold curves and score distribution
analysis_output/weight_sensitivity_report.md	Generated report of the in-sample sensitivity analysis

The software was written in Python, and the Random Forest was built with the scikit-learn library [12]. Running python notebooks/threshold_analysis.py from the service

folder regenerates the held-out results reported in Chapter 6. The analysis script and the data are provided as appendices or in the project repository.

4 PRODUCT AND SOLUTION DESIGN

This chapter describes what the employer legitimacy verification component must do, where it sits in the complete solution, the design decisions that were made and why, and the detailed design of each part. The research method is in Chapter 3, and the implementation, integration, and testing are in Chapter 5.

4.1 Requirements

4.1.1 Stakeholders and users

Three kinds of users interact with the component. Job seekers see its result through the platform, as a risk band with reasons. The recommendation component of the group solution is a system user that consumes the score, band and evidence. The maintainer (the author) needs the component to be configurable, testable and easy to extend with new registry or reputation sources. The external validators review the work and its usefulness.

4.1.2 Functional requirements

Table 4.1 lists the functional requirements. Each is traced to an objective of Section 1.5, and the test cases that cover them are in Chapter 5.

Table 4.1: Functional requirements

ID	Requirement	Objective
FR1	Accept a company name, a website URL and an email address, any of which may be missing, and return “Unknown” with a warning when none is given	O6
FR2	Normalize the company name and URL, recognize job-board links so that they are not treated as the employer’s own site, and find the company website when none is given	O2
FR3	Check Sri Lankan registers (CSE, eROC/DRC, CBSL, BOI, IRCSL and SLAASMB as an extra source) and merge the results from all sources	O2
FR4	Collect reputation signals from job and review platforms, social pages and scam reports	O2

FR5	Analyze the employer website: HTTPS, About, Contact, privacy and terms pages, payment-risk wording and urgency wording	O2
FR6	Compute a model score from the structured features using the trained Random Forest	O3
FR7	Combine the model score and the three-layer scores into a legitimacy score from 0 to 100 and assign a Low, Medium, or High-risk band	O4, O5
FR8	Return an evidence report: registration, reputation and website evidence, the registration trace, the per-layer score breakdown and a recommendation	O2, O6
FR9	Treat a missing registry match as "not confirmed" and not as proof of fraud, and avoid placing a likely genuine SME in the High band for this reason alone	O5
FR10	Report low confidence, with a warning, when too little information is given or the website cannot be scanned	O6
FR11	Expose the component as a service that other components can call and return the result in a structured format	O6

4.1.3 Non-functional requirement

Table 4.2: Non-functional requirements

ID	Requirement	How it is verified
NFR1 Explainability	Every result must be accompanied by the evidence that produced it	Inspection of the response fields; tests in Chapter 5
NFR2 Performance	The result should be returned in a time acceptable for interactive use, so the independent checks run in parallel and results are cached	Response time measured over repeated requests (Chapter 5)
NFR3 Reliability	A failure of one source (timeout, blocked page, changed portal) must not stop the others; the result must degrade to lower confidence instead of failing	Tests with unreachable sources (Chapter 5)

NFR4 Maintainability	Registry, reputation, website, model and scoring logic are kept in separate modules, with settings (timeouts, cache lifetimes) taken from environment variables	Code structure (Table 4.8)
NFR5 Security and privacy	Use public sources only, keep access keys out of source code, store no user data	Section 3.6; code review
NFR6 Interoperability	A documented request and response format over HTTP that the other components can use without code changes	Integration test (Chapter 5)
NFR7 Reproducibility	Training and evaluation can be repeated with fixed seeds and scripts	Section 3.7

4.2 Component architecture within the complete solution

The complete solution has four components (Figure 4.1). The fake job post detector, the employer legitimacy verifier and the scam communication detector each give an independent risk signal, and the recommendation component uses these signals when it ranks jobs. The employer verifier receives the employer details taken from a job advertisement and returns the legitimacy score, the risk band and the evidence trace. It runs as its own service, so that it can be developed, tested and deployed separately from the others.

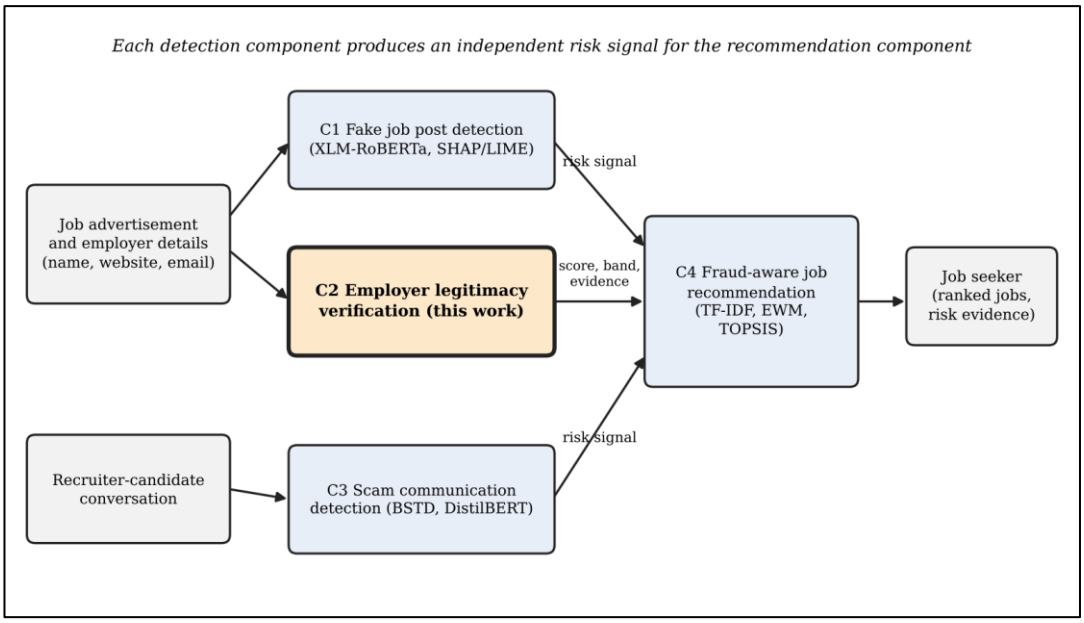


Figure 4.1: Position of the employer verification component in the complete solution

Table 4.3: Interfaces of the component

Interface	**Direction**	**Format**	**Purpose**
POST /predict	In from the client or platform	JSON: company_name, email, website_url	Request a verification
Response of /predict	Out to the client and the recommendation component	JSON: score, band, verdict, evidence, breakdown, recommendation	Return the result and its explanation
GET /	Out	JSON status message	Check that the service is running
Sri Lankan registers (eROC/DRC, CSE, CBSL, BOI, IRCSL, SLAASMB	Out to public sources	HTTP requests, public pages, listings and APIs; headless browser as a fallback	Confirm that the organization exists
Review and job platforms, social pages	Out to public sources	Web search and page requests	Collect reputation signals
Employer website	Out to the website	HTTP requests	Collect website signals
Model bundle	Read at start-up	Serialised model and feature list (joblib file)	Load the trained Random Forest

Figure 4 shows the internal architecture. The request is validated and prepared first. The registry, reputation and website checks are independent of each other and run at the same time. Their results are turned into model features and also passed directly to the scoring layer, which combines them with the model score into the final result. Because each layer works alone, a layer that fails or finds nothing lowers the confidence of the result and does not stop it.

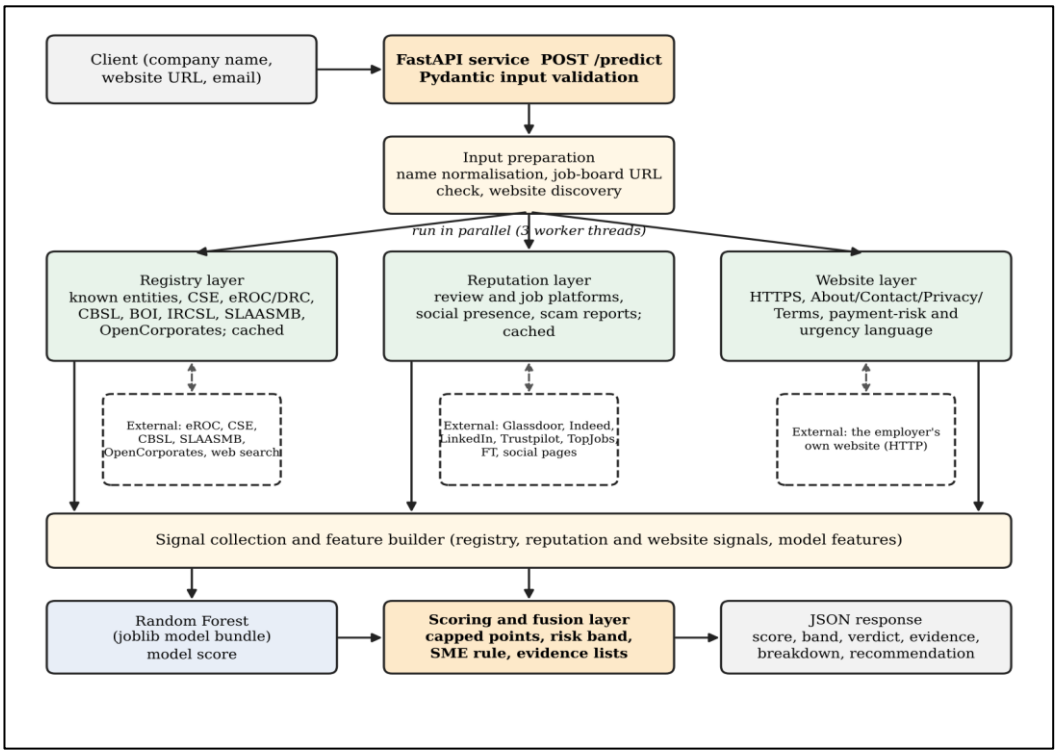


Figure 4.2: Internal architecture of the employer verification service and its external interfaces

4.3 Design alternatives and decisions

Table 4.4 records the main design decisions, the alternatives that were considered and the reasons for the choice. The reasons are design arguments, and the alternatives were not all benchmarked in this work.

Table 4.4: Design decisions

Decision	Alternatives considered	Choice and reason
How to combine evidence	A single classifier on all signals; a learned combination of layers; hand-written rules only	A weighted combination of four-layer scores. Each layer can be explained and the weights can be justified by experiment (Chapter 6). A single classifier hides which evidence mattered, and rules

		alone cannot use the learned patterns.
Model	Logistic regression; gradient boosting; a text model on the website	Random Forest [4] on 14 structured features. It handles mixed binary and count features, needs little tuning, gives a score for each class, and its feature importance can be inspected. A text model would need much more data and would repeat the work of the fake-job component.
Explanation method	SHAP or LIME on the model	An evidence report built from the registry, reputation and website results. The score is already composed of interpretable layers, and the evidence is expressed in terms the job seeker understands (for example, “listed on the CSE”). SHAP and LIME are used in the fake-job component.
Registry sources	One source only; paid commercial databases	Several public sources merged, because no single public Sri Lankan source covers all employers. Paid databases were not used, so that the method can be reproduced without cost.
Missing registry entry	Treat as a strong fraud signal	Treat as “not confirmed”. Many genuine SMEs are not on the CSE or BOI lists, and treating them as fraud would harm genuine employers.
Running the checks	One after another	In parallel with three worker threads, because the three checks are independent and each wait for the network.
Repeated requests	No caching	Short-lived caches: a confirmed registration is kept for longer than a miss, so that a failed lookup does not persist.
Packaging	A library called inside another component; a separate service	A separate HTTP service on its own port, so that the other components, written by other members, can call it without sharing code or libraries.

Table 4.5: Lists the tools and technologies that were used and why

Tool	Use	Justification
Python	Implementation language	Same language as the machine-learning libraries and the other components
FastAPI, Uvicorn, Pydantic	Web service, server and request validation	Fast to build, validates the input automatically and produces interactive documentation [16]
scikit-learn [12], pandas, NumPy	Model training, evaluation and data handling	Standard, well-documented libraries for the Random Forest and the evaluation scripts
joblib	Saving and loading the model with its feature list	The feature order is stored with the model, which prevents a mismatch between training and use
requests, BeautifulSoup	Page requests and HTML analysis	Lightweight and sufficient for static pages
Selenium with a headless browser	Fallback for registry portals that need a browser	Some portals build their results with scripts; it is used only as a fallback because it is slower
duckduckgo-search	Finding the company website and review pages by name	Needs no paid key, so the service can be reproduced
python-whois	Domain information used in feature preparation	Gives domain data for the website analysis
python-dotenv, environment variables	Settings and keys	Keeps keys and timeouts out of the source code

4.4 Detailed design

4.4.1 Service interface

The service has one analysis endpoint, POST /predict, and a status endpoint, GET /. The request has three optional fields. Table 4.6 shows the main response fields.

Table 4.6: Main fields of the /predict response

Field	Content
prediction, probability, confidence, warning	Model decision (Legit, Fake, or Unknown), its score, and a confidence level with a warning when the evidence is thin
risk_score, risk_level, verdict, color	Legitimacy score from 0 to 100, the band (Low, Medium or High), a verdict sentence and a display color

evidence	Registration, reputation and website evidence lists, the per-feature evidence, and the positive, negative and unknown features
registration_status, registration_trace, registration_sources, reg_name, reg_number	Result of the registry checks, the step-by-step trace of sources tried, and the source and number of the match
score_breakdown	Points from the model, registration, reputation and website layers
recommendation	Advice for the job seeker for the band
features_used, elapsed_seconds	The model inputs and the time taken

4.4.2 Processing sequence

1. The request is validated and the company name and URL are normalized. A job-board link is recognized, and a company website is searched for by name when none is given.
2. The registry, reputation and website checks run in parallel.
3. The results are converted into the model features, and the Random Forest gives the model score.
4. The scoring layer awards points for each layer, adds them, applies the SME rule and assigns the band.
5. The confidence rules are applied, and the response is built with the evidence report.

4.4.3 Registry layer

The registry layer first expands brand names to legal names with an alias table (for example, a trade name to the registered company). It then queries the sources and merges what each return, without stopping at the first hit: a curated list of known registered entities, the CSE listings, the eROC/DRC search, searches for CBSL, BOI and IRCSL records, and the SLAASMB list of Specified Business Enterprises. Every step is written to a registration trace with its source, status and detail, so that the result can be checked. Positive results are cached for 30 minutes and misses for at most two minutes, and the CSE listing is cached for an hour. The SLAASMB list is cached for a week with a stored seed list as a fallback when the live page cannot be read.

4.4.4 Reputation layer

The reputation layer looks for the company on Glassdoor, Indeed, LinkedIn, Trustpilot, TopJobs.lk, Daily FT, ikman.lk and social platforms, using links on the company website, the given URL and a search by company name. It also looks for review text with positive or negative wording and for scam reports. A company that appears only on social media is flagged as social-only presence, which the scoring layer treats as a smaller footprint. A scam report is treated as a hard negative signal. Results are cached for 30 minutes.

4.4.5 Website layer and features

The website layer fetches the page with a short timeout, finds the About, Contact, privacy and terms pages from the links and from common paths, and measures the use of HTTPS, the number of subdomain levels, payment-risk wording (requests for fees), urgency wording and a content score. A headless browser is used only when enabled by a setting. Job-board pages are not used as the employer's own website. These signals, the registry result and the review-platform presence form the 14 features of the Random Forest (Table 3.2).

4.4.6 Model component

The Random Forest is trained offline on the training split and saved with its feature list as a single file that the service loads at start-up. For each request the features are placed in the stored order and the model returns a score for the class "legitimate", called the model score. The model is not retrained by the service.

4.4.7 Scoring and fusion layer

The scoring layer expresses the four-part weighting of Section 3.3 as capped points, so that each layer has a visible maximum: the model score gives up to 40 points, registration up to 30, reputation up to 20 and website signals up to 10. Table 4.7 shows how the registration points are awarded and how the bands are set.

Table 4.7: Scoring rules of the deployed layer

Part	Rule
Model score (up to 40)	Model score × 40

Registration (up to 30)	The highest of: CSE listing 30; BOI, CBSL, IRCSL or SLAASMB 25; DRC/eROC 20; other government registration 15
Unconfirmed registration	An unverified registration hint gives 10. A business that is not registered on the lists but has a website, a social or review footprint, no scam signal and a model score of at least 0.55 gets 8 or 12 points and is described as "common for SMEs". Otherwise, the layer gives 0 and states that no registration was found.
Reputation (up to 20)	Points for each platform found (Glassdoor, Indeed and LinkedIn 5 each; TopJobs 3; Daily FT, Trustpilot, Sitejabber and ikman 2 each), for social presence (1 point each, 3 when social media is the main footprint) and for positive reviews (4). Negative reviews subtract 4 and a scam report subtracts 15. The total is kept between 0 and 20.
Website (up to 10)	HTTPS 3; About page 2; Contact page 2; privacy or terms page 2; 2 points are subtracted for weak security signals. A business whose public presence is only on social media receives partial credit (3 to 5).
Risk bands	Score of 70 or more: Low. 45 to 69: Medium. Below 45: High.
SME rule	If the score is below 45, there is no payment-risk or scam signal, and the registration is not confirmed, a model score of at least 0.65 (or a company-like name with a score of at least 0.45 and a website or review footprint) raises the score to 48, which keeps the employer in the Medium band, with a note asking for the registration number.

The evaluation of Chapter 6 uses the normalized formula with the weights 0.40, 0.30, 0.20 and 0.10, while the deployed layer uses the capped points above, with graded registry points and the SME rule. The differences, and what they mean for the results, are discussed in Chapters 5 and 7.

4.4.8 Evidence report and confidence

The evidence report is the explanation the component gives. It lists the registration evidence (for example "Listed on the Colombo Stock Exchange"), the reputation evidence (for example "Found on LinkedIn"), the website evidence, and the positive, negative and unknown features. The score breakdown shows how many points each layer contributed, and a list of confidence reasons gives the main points in plain

language. The confidence is set to low, and the model score reported is limited to 0.65, when only one input is given, when the website could not be scanned, or when at least four of six content signals are missing. When no input is given the result is "Unknown".

4.4.9 Code structure

Table 4.8: Code structure of the service

File or folder	Content
app/main.py	Service, request model, feature preparation, parallel checks and response
app/employer_verification_model/registration_utils.py	Registry checks, source merging, registration trace and caches
.../slaasmb_lookup.py, known_registered.py, company_aliases.py	SLAASMB list, curated known entities and alias resolution
.../review_aggregator.py	Reputation and review signals
.../feature_engine.py	Website fetching and website signals
.../scoring_layer.py	Points, bands, SME rule, evidence and recommendation
.../final_realistic_model.py	Model training and saving
notebooks/, app/weight_sensitivity_analysis.py	Dataset preparation, threshold and weight analysis (Chapter 3)
tests/ and test scripts	Tests of the scoring layer and of the full service (Chapter 5)

4.5 Standards and Professional Practice

4.5.1 Interface standards

The service follows the REST style [13], using HTTP and JSON with a typed request model, and its interactive documentation is generated from the code, which helps other members use it.

4.5.2 Data protection

Sri Lanka's Personal Data Protection Act, No. 9 of 2022 [14], governs the processing of personal data. The component works with organization-level information from public sources, stores no user data, and does not keep request history. The email

address, when given, is used only to derive features for that request. [Confirm with your supervisor whether the data-protection or ethics review of the faculty applies.]

4.5.3 Professional ethics

The design follows the principle of avoiding harm in the ACM Code of Ethics [15]. A wrong result about a named company can harm a genuine business, so the output is a risk estimate with reasons and not an accusation; a missing registration is “not confirmed”, and the recommendation asks the job seeker to verify with the employer through official channels.

4.5.4 Engineering practice

The code is divided into modules with one responsibility each, settings are read from environment variables, keys are not stored in the source code, the work is kept under version control, and the scoring and the full service have test scripts (Chapter 5). External sources are accessed only for public information, and the number of requests per lookup is limited by timeouts and caching. [Add any further rate-limiting or politeness measures that you implemented.] The use of AI tools in preparing this dissertation is disclosed in Appendix B.

5 IMPLEMENTATION INTEGRATION AND TESTING

This chapter describes how the design of Chapter 4 was built, how the component connects to the rest of the solution, and how it was tested. Section 5.1 gives the environment, Section 5.2 the implementation of each part, Section 5.3 the integration with the other components, Section 5.4 the tests and their results, and Section 5.5 the deployment.

5.1 Development environment and technology stack

The component is written in Python and runs as a web service. Table 5.1 lists the environment. The libraries are those of Table 4.5, and the code is kept under version control. Keys and timeouts are read from environment variables and are not written in the source code.

Table 5.1: Development environment

Item	Used
Language and runtime	Python 3.12.10
Service framework	FastAPI with Uvicorn; request model checked by Pydantic
Machine learning and data	scikit-learn 1.9.0, pandas, NumPy, joblib
Web access	requests, BeautifulSoup, duckduckgo-search, python-whois; Selenium with a headless browser as a fallback
Testing	pytest 9.1.1
Operating system and editor	Win32NT 10.0.19045.0 Microsoft Windows NT 10.0.19045.0
Code repository	https://github.com/Kaju2002/R26-SE-002.git

5.2 Component implementation

The design of each layer, the scoring rules and the evidence report are described in Section 4.4 and are not repeated here. This section records how that design was turned into code: the order in which the service works, how the time and the settings are controlled, and some rules that are written directly in the code.

5.2.1 Order of work in the service

The service is written as a FastAPI application whose analysis endpoint, POST /predict, calls one function, extract_features_from_input, to build the model features, and then the scoring function. The code follows the processing sequence of Section 4.4.2. The company name and address are normalized first, and a job-board link is

recognized so that it is not used as the employer's own site (FR2). When no website is given, the function searches for one by company name. If all three request fields are empty, the service returns the result Unknown with a warning (FR1).

The registry, reputation and website steps are submitted to a pool of three worker threads (ThreadPoolExecutor with three workers), because each of them mostly waits for the network. The registry and reputation results are returned together with the features, and the scoring function accepts them as inputs, so that no source is queried a second time for the same request. The website step is wrapped so that an error returns empty signals and the request continues with a lower-confidence result (FR10, NFR3). The time of the parallel stage is written to the log, and the total time is returned in the field elapsed_seconds.

5.2.2 Time limits and settings

Every call to an outside source has a time limit, so that one slow source cannot hold the whole request. The limits and the cache lifetimes are read from environment variables with the defaults in Table 5.2, so that they can be changed without editing the code (NFR4). The headless browser is off by default and is switched on by a setting only for the portals that need it. The registry and reputation results are kept in memory with a timestamp, and the cache lifetimes of Section 4.4.3 are applied by comparing that timestamp with the stored limit.

Table 5.2: Main settings of the service (environment variables and defaults)

Setting	Default	Purpose
WEBSITE_HTTP_TIMEOUT, WEBSITE_PROBE_TIMEOUT	5 s, 3 s	Time limit when fetching a company page, and when probing common paths
REG_HTTP_TIMEOUT	5 s	Time limit for registry page requests
CSE_API_TIMEOUT, CSE_CACHE_TTL_SEC	6 s, 3600 s	Time limit for the CSE listing request, and how long the listing is kept
EROC_SELENIUM_TIMEOUT	18 s	Time limit for the headless eROC search
SLAASMB_HTTP_TIMEOUT, SLAASMB_CACHE_TTL_SEC	12 s, 7 days	Time limit for the SLAASMB page, and how long its list is kept (a stored seed list is the fallback)
REG_RESULT_CACHE_TTL_SEC	1800 s	How long a confirmed registration result is kept; a miss is kept for at most 120 s

REP_RESULT_CACHE_TTL_SEC	1800 s	How long a reputation result is kept
WEBSITE_USE_HEADLESS, EROC_USE_SELENIUM, CSE_USE_SELENIUM	Off	Switch the headless browser on for the website, eROC and CSE steps
OPEN_CORPORATES_API_TOKEN	Not set	Optional token for the OpenCorporates source

5.2.3 Rules written in the code

Some rules are keyword lists in the code. The website text is checked against a list of payment phrases (for example, cryptocurrency, wire transfer, Western Union, send money first, pay to apply) to set the payment-risk flag, and against a list of pressure phrases (for example, act now, limited time, apply immediately, last chance) to set the urgency flag. A separate count of twelve scam phrases, such as registration fee and guaranteed job, is also kept. The reputation step marks a scam report when the company name itself contains a word such as scam, fake, fraud or easy money. These lists are simple and can be evaded or can match honest pages, so the flags are treated as evidence in the report and not as a verdict. The lists can be extended in one place.

5.2.4 What the implementation reuses

The model, the scoring rules and the evidence report are produced by the modules listed in Table 4.8. The model file is loaded once at start-up with its feature list, and the features are placed in the stored order for each request. The scoring function builds the evidence report from the same points that give the score, so the report cannot disagree with the number (NFR1).

5.3 Integration with the other components

The component runs as its own service on its own port (8001 in the start script, while the scam-detection service uses port 8000), so that the other members can call it without sharing code or libraries (NFR6). The mobile application, FraudAware, sends the employer's name and website taken from a job advertisement to POST /predict and shows the band, the verdict and the evidence report to the job seeker.

The recommendation component receives one risk value from this component. The employer risk is $r = 1 - S/100$, where S is the legitimacy score from 0 to 100, so a score of 70 or more gives a risk of 0.30 or less. The risk is stored with the job together

with the evidence report and the risks of the other two detectors, and the recommendation component uses the three risks when it ranks jobs. The interface is the response described in Table 4.6, and the other components need only the score, the band and the evidence fields.

The hand-over was checked by calling the service from the mobile application / a test client and reading the fields of Table 4.6 and it is passing

5.4 Verification and software testing

5.4.1 Test approach

Two kinds of tests were used. Automated unit tests check the rules that can be tested without network access: the scoring layer, with registry and reputation inputs supplied directly, and the saved model bundle. Scenario tests call the service through its HTTP interface with fixed inputs in place of the public sources, because the answers of live sources change from day to day and a test on them would pass or fail by chance. The automated tests are in the tests folder of the service and are run with the command python -m pytest tests.

5.4.2 Automated unit tests

Table 5.3 lists the 38 automated tests by group and the requirement each group verifies. All 38 tests passed. The scoring-layer tests (31) take the point rules of Table 4.7 and check that each rule gives the stated number of points, that the caps hold, that the highest registry source counts instead of the sum, that the bands start at the stated boundaries, and that a company with a missing register entry and a good footprint is not placed in the High band. The model tests (7) check that the saved bundle has the 14 features and the forest settings reported in Chapter 3, that the score is between 0 and 1, that the class legitimate is label 1, that a clean profile scores higher than a scam profile, and that retraining on the training split with the fixed seed gives a test accuracy above 0.90 on the 300 test employers.

Table 5.3: Automated tests and their results

Group	Tests	Verifies	Result
Registration points (CSE 30; BOI, CBSL, IRCSL, SLAASMB 25; DRC 20; other government 15; unverified 10; highest source counts)	9	FR3, FR7	9 passed

Reputation points (platforms, cap of 20, scam report, negative reviews)	5	FR4, FR7	5 passed
Website points (HTTPS, About, Contact, privacy or terms, weak security, cap of 10)	3	FR5, FR7	3 passed
Model part, risk bands and score range	5	FR6, FR7	5 passed
Missing registry entry and the SME rule	6	FR9	6 passed
Evidence reports fields and breakdown	3	FR8, NFR1	3 passed
Saved model bundle and held-out accuracy	7	FR6, NFR7	7 passed
Total	38		38 passed

The tests check the scoring layer against the rules written in this thesis, so a change in the code that breaks a rule is detected. They do not prove that the rules are the best possible rules; that is the purpose of the evaluation in Chapter 6.

5.4.3 Scenario tests of the service

The behavior of the whole service, which the unit tests do not cover, was tested with scenarios. Table 5.4 lists the eight cases and their results. The cases were run through the HTTP interface of the service with the model and the scoring layer unchanged. The calls to the registry, review and website sources were replaced by fixed inputs, so that each case was repeatable and the answers did not depend on the state of the public sources on the day. Live access to the public sources was therefore not part of these tests; it is covered by the evaluation of Chapter 6, which used features collected from the live sources, and it remains the part of the service that depends on the network.

Table 5.4: Scenario tests of the service and their results

Test	Input	Expected result	Requirement	Result
No input	All three fields empty	Unknown with a warning, and no band that can be mistaken for a decision	FR1	Prediction Unknown with a warning, but the band was Medium (48). Defect D1
Job-board link	A TopJobs link as the website	The link is not treated as the employer's site	FR2	Passed: no site features used, low confidence, Medium (48)
Listed company	A company listed on the CSE, with LinkedIn, Glassdoor and Facebook pages	Low band, CSE evidence in the report	FR3, FR7	Passed: Low (83.8), registered, high confidence
Unlisted small company with a clean site	A small company with a complete	Medium band, registration	FR9	Passed: Medium (46.5), "Not listed on CSE/BOI

	website and a LinkedIn page	described as not confirmed		(common for SMEs)”
Website cannot be scanned	Same company, website returns no content	Result returned with a lower-confidence flag	FR10, NFR3	Passed: Medium (56.9), confidence low, warning shown
Scam signals	No registry match, scam report, payment wording, thin site on a xyz domain	High band	FR6, FR7	Passed: High (35.4), confidence low. The field prediction read Legit (Defect D3)
One source fails	The registry call raises an error	Other checks still return; confidence lowered	NFR3	Failed: HTTP 500 and no result. Defect D2
Response time and interface	20 repeated requests; fields of Table 4.6	Fast answer; all fields present	NFR2, FR11, NFR6	Passed: median 4 ms, maximum 9 ms with the sources replaced; all fields present

5.4.4 Findings from testing

The saved model was written with scikit-learn 1.9.0, and loading it with 1.9.1 gives a version warning. The results were not affected, but the requirements file does not fix the library versions. The versions used should be recorded in the requirements file so that the model can be loaded in the same way later (Section 5.5).

The scenario tests found three defects. They were not corrected in the version evaluated in this thesis because the evaluation of Chapter 6 was run on that version, and they are reported here with the correction that each one needs.

- D1: an empty request returns the band Medium with a score of 48, although the prediction field says Unknown. The routing rule of Section 5.3.2 would verify such a request automatically. The correction is to return no band for an Unknown result and to send it to the administrator.
- D2: if a registry call raises an error that is not handled inside the registry module, the service answers with HTTP 500 and no result, which breaks NFR3. The correction is to catch the error around each of the three parallel calls and continue with an empty result and the lower-confidence flag. Errors that the registry module itself handles, for example a timeout of one source, were not tested here.
- D3: the field prediction, which is the output of the Random Forest on its own, can read Legit while the band is High, as in the scam case of Table 5.4. The band is the

field that the platform should use, and the field prediction should be labelled as the model part only or removed from the response.

The response time of 4 ms measures the model and the scoring layer only. The time of a live check is dominated by the three parallel source calls and by the timeouts of Table 5.2, and it was not measured in these tests.

5.5 Deployment and operation

The service is started with Uvicorn on port 8001 using the script run.ps1, and its interactive documentation is generated from the code. It reads its settings from environment variables, and the model file is loaded once at start-up. Each response contains the time taken, so that response times can be monitored. Repeated requests for the same company use the short-lived caches described in Section 4.4.3. For the project, the service is started locally with run.ps1 and is reached by the other components over HTTP on its own port; the same start command and settings would be used on a server, and only the address in the other components would change.

Three points matter for operation. The registry and review sources are public pages that may change or limit automated access, so a layer can fail and the result then carries a lower-confidence flag. The library versions should be fixed in the requirements file, because the saved model depends on the scikit-learn version. Finally, the component stores no user data and keeps no request history, as required by Section 4.5.2. The service sets no request rate limit of its own. The load on the public sources is limited by the pool of three workers, the timeouts and the caches, and a deployment with many registrations would need a request limit in front of it.

6 RESULTS AND VALIDATION

This chapter reports what the experiments showed, in the order of the research questions of Section 1.6. All results in Sections 6.1 and 6.5 come from the held-out evaluation described in Chapter 3: the model was trained on the training split only, weights and thresholds were chosen on the validation split, and the frozen designs were evaluated once on the test split of 300 employers (195 legitimate, 105 fraudulent). Fraud recall is the share of fraudulent employers scored below the cut-off, and the false positive rate (FPR) is the share of legitimate employers wrongly flagged, as defined in Table 3.3. Intervals are Wilson 95% intervals. The chapter reports what happened; Chapter 7 explains why.

6.1 Research results

6.1.1 Random Forest on the structured features (RQ2, O3)

Table 6.1 shows the Random Forest at the cut-off of 0.5 on the model score. Legitimate is the positive class. The accuracy is 0.926 on the training split, 0.923 on the validation split and 0.917 on the test split, so the gap between training and test is small and the model does not appear to be overfitted. On the test split, the model finds 179 of the 195 legitimate employers and 96 of the 105 fraudulent ones. The area under the ROC curve is 0.973 on the test split (95% bootstrap interval 0.954 to 0.987).

Table 6.1: Random Forest results by split (cut-off 0.5 on the model score)

Split	n	Accuracy	Precision	Recall	F1	AUC	Fraud recall	FPR
Training	901	0.926	0.955	0.927	0.941	0.981	0.923	0.073
Validation	313	0.923	0.941	0.941	0.941	0.982	0.889	0.059
Test	300	0.917	0.952	0.918	0.935	0.973	0.914	0.082

Precision, recall and F1 describe the legitimate class. On the test split the model misses 9 fraudulent employers (scored as legitimate) and flags 16 genuine employers as fraudulent. These are the results of the model alone. The fused score, which adds registry, reputation and website evidence, is reported next.

6.1.2 Fusion weights and the price of explainability (RQ3, O4)

Five weight profiles and a grid search were compared. Table 6.2 gives the results on the validation split, which was used for the choice, and on the test split, which was used once. The proposed weights (0.40, 0.30, 0.20, 0.10) reach an accuracy of 0.870

and a fraud recall of 0.781 on the test split. The weights found by the grid search on the validation split (0.50, 0.10, 0.25, 0.15) reach 0.930 and 0.933. The ML-dominant weights reach 0.920 and 0.876.

Table 6.2: Weight profiles on the validation and test splits (cut-off 45)

Profile (weights)	Split	Accuracy	Precision	Recall	F1	Fraud recall	FPR
Proposed (.40/.30/.20/.10)	Validation	0.847	0.869	0.902	0.885	0.741	0.098
	Test	0.870	0.886	0.918	0.902	0.781	0.082
ML-dominant (.70/.10/.10/.10)	Validation	0.936	0.938	0.966	0.952	0.880	0.034
	Test	0.920	0.934	0.944	0.939	0.876	0.056
Registry-heavy (.20/.60/.10/.10)	Validation	0.741	0.754	0.898	0.820	0.444	0.102
	Test	0.733	0.745	0.897	0.814	0.429	0.103
Equal (.25 each)	Validation	0.840	0.860	0.902	0.881	0.722	0.098
	Test	0.850	0.868	0.908	0.887	0.743	0.092
Reputation ignored (.50/.40/0/.10)	Validation	0.789	0.768	0.971	0.858	0.444	0.029
	Test	0.773	0.757	0.959	0.846	0.429	0.041
Grid best (.50/.10/.25/.15)	Validation	0.952	0.966	0.961	0.963	0.935	0.039
	Test	0.930	0.963	0.928	0.945	0.933	0.072

Table 6.3 adds the Wilson intervals on the test split, and the Random Forest alone as a reference. The intervals are wide because the test split has 105 fraudulent employers. The proposed weights miss 22% of the fraud, against 6.7% for the grid-search weights and 8.6% for the Random Forest alone, and the false positive rates of the three are close (0.082, 0.072 and 0.082). The price of choosing weights that every layer visibly shares is therefore about 15 points of fraud recall against the best-tuned weights, and about 13 points against the model alone, at a similar false positive rate.

Table 6.3: Test results with Wilson 95% intervals (n = 300)

Profile	Accuracy	Fraud recall [95% interval]	FPR [95% interval]
Proposed (.40/.30/.20/.10)	0.870	0.781 [0.693, 0.849]	0.082 [0.051, 0.129]
ML-dominant (.70/.10/.10/.10)	0.920	0.876 [0.800, 0.926]	0.056 [0.032, 0.098]
Registry-heavy (.20/.60/.10/.10)	0.733	0.429 [0.338, 0.524]	0.103 [0.067, 0.153]
Equal (.25 each)	0.850	0.743 [0.652, 0.817]	0.092 [0.059, 0.141]
Reputation ignored (.50/.40/0/.10)	0.773	0.429 [0.338, 0.524]	0.041 [0.021, 0.079]
Grid best (.50/.10/.25/.15)	0.930	0.933 [0.869, 0.967]	0.072 [0.043, 0.117]

Random Forest alone (score < 0.5)	0.917	0.914 [0.845, 0.954]	0.82 .051, 0.129]

6.1.3 What each evidence source adds (RQ3)

Table 6.4 gives the area under the ROC curve of each source alone and of the fused score with one source removed. The model score is the strongest single source (AUC 0.973) and reputation follows (0.911). The registry score alone separates the classes poorly (0.661), and the website score alone is close to chance on the primary data (0.607). Removing the model from the proposed fusion lowers the AUC from 0.951 to 0.873, and removing reputation lowers it to 0.928. Removing the registry layer raises the AUC to 0.985, and removing the website layer leaves it almost unchanged at 0.955.

Table 6.4: AUC on the test split by evidence source (primary data)

Score	AUC [95% bootstrap interval]
Model score only	0.973 [0.954, 0.987]
Registry only	0.661 [0.593, 0.732]
Reputation only	0.911 [0.877, 0.937]
Website only	0.607 [0.544, 0.670]
Proposed fusion	0.951 [0.930, 0.971]
Grid-best fusion	0.983 [0.969, 0.994]
Fusion without model	0.873 [0.828, 0.913]
Fusion without registry	0.985 [0.973, 0.995]
Fusion without reputation	0.928 [0.897, 0.954]
Fusion without website	0.955.933, 0.973]

The importance of the features in the forest points the same way. On the primary data, the content score (0.34), the payment-risk flag (0.29) and the urgency flag (0.15) carry most of the weight, and registration status follows with 0.11. The three review-platform features together carry less than 0.01.

6.1.4 Thresholds and risk bands (RQ4, O5)

Table 6.5 shows the sweep of the cut-off on the validation split. Fraud recall rises from 0.426 at a cut-off of 20 to 0.981 at 65, while the false positive rate rises from 0 to 0.215. The weighted error cost has its minimum at a cut-off of 65 for all three cost ratios (3:1, 5:1 and 10:1). Above 70, the false positive rate grows quickly (0.449 at 70 and 0.795 at 75).

Table 6.5: Threshold sweep on the validation split (selected rows)

Cut-off	Fraud recall	FPR	Legitimate precision at or above cut-off	Cost 3:1	Cost 5:1	Cost 10:1
40	0.648	0.039	0.838	122	198	388
45	0.741	0.098	0.869	104	160	300
50	0.852	0.112	0.919	71	103	183
55	0.907	0.127	0.947	56	76	126
60	0.944	0.161	0.966	51	63	93
65	0.981	0.215	0.988	50	54	64
70	1.000	0.449	1.000	92	92	92

Table 6.6 gives the risk bands on the test split for two sets of boundaries. The deployed boundaries (45 and 70) were set before the experiments. The other pair (55 and 60) was selected by the two rules on the validation split described in Section 3.3. With the deployed boundaries none of the 107 employers in the Low band is fraudulent. The High band holds 82 of the 105 fraudulent employers (78.1%, Wilson interval 69.3% to 84.9%), together with 16 legitimate employers, and 72 legitimate employers fall in the Medium band. With the validation-selected boundaries the High band holds 96 of 105 fraudulent employers (91.4%, interval 84.5% to 95.4%) and 29 legitimate employers, but 5 fraudulent employers reach the Low band.

Table 6.6: Risk bands on the test split (number of legitimate/fraudulent employers)

Boundaries	High	Medium	Low	Fraud in High	Fraud in Low
Deployed 45 / 70	16 / 82	72 / 23	107 / 0	82 of 105 (78.1%)	0 of 107
Validation-selected 55 / 60	29 / 96	13 / 4	153 / 5	96 of 105 (91.4%)	5 of 158

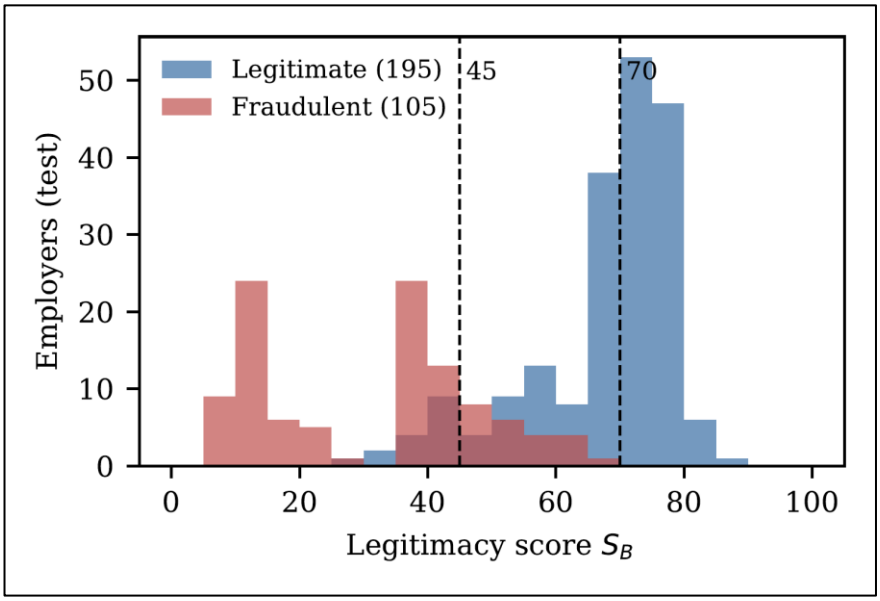


Figure 6.1: Distribution of the fused score on the test split, with the deployed bands at 45 and 70

Figure 6.1 shows the score distribution. The mean score is 67.5 for legitimate employers and 31.8 for fraudulent ones, and the two classes overlap between about 35 and 60, which is where the Medium band lies.

Figure 6.2 shows how fraud recall and the false positive rate change with the cut-off on the validation split, and so how the bands of Table 6.6 were chosen.

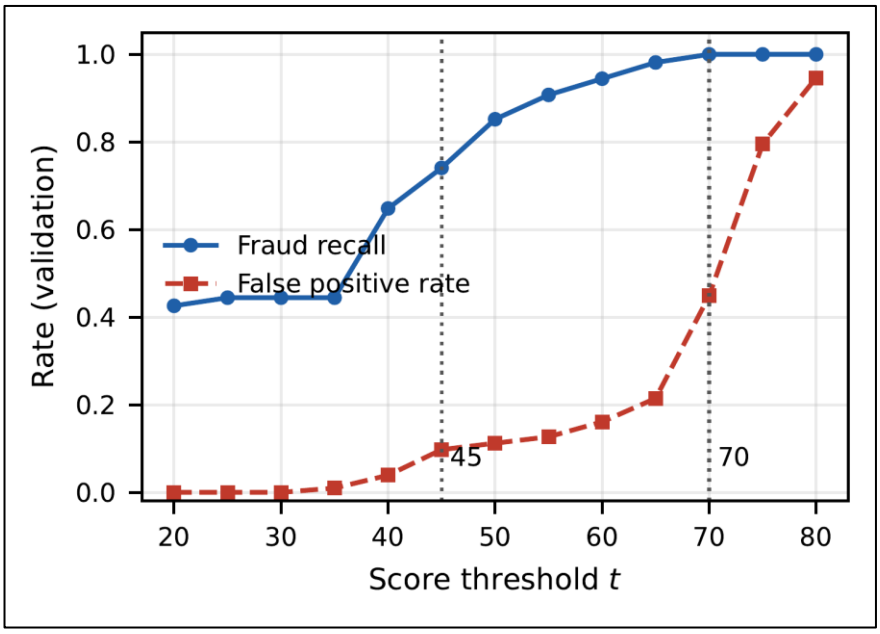


Figure 6.2: Fraud recall and false positive rate against the cut-off on the validation split

6.1.5 Error analysis

Table 6.7 gives the mean of each score by class on the test split. The model and reputation scores separate the classes most (0.79 against 0.21 for the model score). The registry score separates them partly (0.86 against 0.58), and the website score hardly at all (0.64 against 0.60).

Table 6.7: Mean score by class on the test split

Class	Model score	Registry	Reputation	Website	Fused score
Legitimate (195)	0.79	0.86	0.19	0.64	67.5
Fraudulent (105)	0.21	0.58	0.01	0.60	31.8

With the deployed cut-off of 45, the component misses 23 fraudulent employers and flags 16 legitimate ones, and the two groups look different. All 23 missed employers are registered; their mean model score is 0.47, and only 13% of them raise a payment-risk flag, against 65% of all fraudulent employers. They are registered companies whose websites look ordinary. None of the 16 flagged legitimate employers is registered, none is present on a review platform, and 25% lack HTTPS. They are small businesses with little public footprint, which is the case that the missing-registry rule and the lower-confidence flag are meant to soften.

6.1.6 Robustness to the data version

The main results use the primary feature table, in which some feature values were randomized at fixed rates to create overlap between the classes (Section 3.2.2). Table 6.8 repeats the main comparison on the as-collected table. The ordering of the designs does not change: the registry-heavy and reputation-ignored profiles remain the weakest, and the Random Forest alone and the grid-search weights remain the strongest. All results are higher on this easier table. With the deployed boundaries, the Low band again contains no fraudulent employers (135 legitimate, 0 fraudulent). With the validation-selected boundaries, it contains 1 (167 legitimate, 1 fraudulent)

Table 6.8: Test results on the as-collected feature table (cut-off 45, n = 300)

Profile	Accuracy	Fraud recall	FPR
Proposed (.40/.30/.20/.10)	0.910	0.914	0.092
ML-dominant (.70/.10/.10/.10)	0.970	0.952	0.021
Registry-heavy (.20/.60/.10/.10)	0.770	0.533	0.103

Equal (.25 each)	0.913	0.924	0.092
Reputation ignored (.50/.40/0/.10)	0.867	0.743	0.067
Grid best (.50/.10/.25/.15)	0.943	0.981	0.077
Random Forest alone (score < 0.5)	0.960	0.971	0.046

6.1.7 Hypotheses

Table 6.9 states whether each working hypothesis of Section 1.6.2 was supported, using the rule of Section 3.5: the held-out result must point in the stated direction, and the difference must be larger than the width of the interval where one is reported.

Table 6.9: Outcome of the hypotheses

Hypothesis	Evidence	Outcome
H1: The proposed weights beat equal weights on held-out data	Test accuracy 0.870 against 0.850, fraud recall 0.781 against 0.743. The intervals overlap widely.	Direction supported; difference not significant on this test split
H2: No single source is sufficient	Registry-heavy and reputation-ignored profiles reach fraud recall of 0.429 against 0.781. The intervals (0.338 to 0.524 and 0.693 to 0.849) do not overlap.	Supported
H3: With bands 45 and 70, few or no fraud reach Low	0 of 107 employers in the Low band are fraudulent (upper 95% limit about 3.5%).	Supported on the test split

6.2 Product and technical results

The component is exposed as a service that accepts a company name and a website address and returns the legitimacy score, the risk band, the evidence report and a confidence flag (Table 4.6). The evidence report lists the registry matches, the platforms found, the website findings, the points of each layer and plain-language reasons, so each statement can be checked on the official source. For an unlisted small company with a clean website, for example, the band is Medium; the report states that no registry match was found and asks the job seeker to request a registration number, and the confidence is lower because only one review platform was found.

Table 6.10 gives the technical results from the tests of Chapter 5. All 38 automated tests passed. Of the eight scenario tests, six passed and two exposed defects (D1 and D2 in Section 5.4.4). When the website could not be scanned, the service returned a result with a lower-confidence flag, but when the registry call raised an unhandled

error, the service returned no result, so the requirement that one failing source must not stop the others is met for the website layer and not yet for the whole service.

Table 6.10: Technical results of the service

Measure	Result	Source
Test cases passed	38 of 38 automated; 6 of 8 scenario tests	Chapter 5, Section 5.4
Response time per request, median (range)	4 ms (4 to 9 ms) over 20 requests, sources replaced by fixed inputs	Chapter 5, Section 5.4
Requests answered with a lower-confidence flag when a source failed	Website not scanned: 1 of 1. Registry call raising an error: 0 of 1	Chapter 5, Section 5.4
Registry sources covered	CSE, eROC/DRC, CBSL, BOI, IRCSL, SLAASMB	Chapter 4

6.3 User or stakeholder validation

No study with job seekers or other users was carried out for this component. The research question concerns how well evidence can be combined and thresholder, which is answered by the held-out experiment of Section 6.1, and a user study would have needed ethical clearance and a working release of the whole solution that was not available during the project period. The product was therefore validated technically (Sections 6.2 and 6.4) and not by users. Whether job seekers understand the evidence report, and whether administrators find the High-band cases useful, remain untested and are listed as limitations in Section 7.5 and as future work in Section 8.4.

6.4 Integration results

The component hands one risk value to the recommendation component. The employer risk is $r = 1 - S/100$, where S is the legitimacy score, so a score of 70 or more gives a risk of 0.30 or less. The evidence report is passed on with the risk so that the recommendation component can show it next to the ranked job. The integration was evaluated in two parts. The interface test of Section 5.3.3 showed that the score, band, confidence and evidence report are present in the response and that the conversion gives higher risk for weaker companies (0.162, 0.535 and 0.646 for scores of 83.8, 46.5 and 35.4).

The second part is the routing of registrations (Section 5.3.2), which was simulated on the 300 test employers using the deployed bands. The High band, 98 employers, would

go to an administrator; it contains 82 of the 105 fraudulent employers (78.1%) and 16 of the 195 genuine ones (8.2%). The other 202 employers, in the Low and Medium bands, would be verified automatically; 179 of them are genuine and 23 are fraudulent. The administrator therefore handles about one third of the registrations (98 of 300) and stops about four in five of the fraudulent ones, and the 23 fraudulent employers that pass automatically are all in the Medium band, none in the Low band. A complete run through the mobile application and the recommendation component was not carried out (Section 7.5).

6.5 Summary of findings

Table 6.11 summarizes the findings against the research questions.

Table 6.11: Summary of findings

Question	Finding
RQ1: Which signals can be collected and turned into features?	Fourteen features from the website, registry and reputation layers were collected for 1,514 employers. Content score, payment-risk and urgency flags and registration status carry most of the model weight.
RQ2: How well does the Random Forest separate the classes on unseen data?	On the test split it reaches an accuracy of 0.917, a fraud recall of 0.914, a false positive rate of 0.082 and an AUC of 0.973. The gap to the training split is small.
RQ3: How should the weights be set, and what does explainability cost?	The proposed weights reach a fraud recall of 0.781. The grid-search weights reach 0.933 and the model alone 0.914, at a similar false positive rate, so the cost of the shared weights is about 13 to 15 points of recall. Registry-heavy weights and ignoring reputation both lose more than half of the fraud (recall 0.429).
RQ4: Do the bands keep fraud out of Low, and at what cost?	With bands 45 and 70, the Low band has no fraudulent employers (0 of 107), the High band holds 78.1% of the fraud, and 72 legitimate employers fall in Medium. Bands 55 and 60 catch 91.4% but let 5 frauds into Low.

7 DISCUSSION

This chapter explains what the results of Chapter 6 mean. Section 7.1 interprets the findings, Section 7.2 compares them with the studies and tools of Chapter 2, Sections 7.3 and 7.4 state the research and product contributions, Section 7.5 sets out the limitations and the threats to validity, and Section 7.6 gives the implications.

7.1 Interpretation of findings

7.1.1 The model learns from the combined evidence (RQ2)

The Random Forest separates the two classes well on employers it has not seen. Its accuracy is 0.926 on the training split, 0.923 on the validation split and 0.917 on the test split, and its AUC is 0.973 on the test split. The small gap between the training and the test split shows that the forest of depth 8 with leaves of at least 8 employers did not memorize the training data. The split was grouped by website domain, so the test result is not raised by near-duplicate records of the same site. The high AUC and the lower fraud recall at the cut-off of 0.5 (0.914) together show that the ranking of employers is better than the single cut-off suggests, which is the reason for the threshold study of Section 6.1.4.

This result must be read with the nature of the data. The feature values of the fraudulent class were partly generated so that the class resembles active scam websites with controlled overlap (Section 3.2.2). The accuracy is therefore evidence that the method works when the signals behave as designed, and it is not an estimate of the accuracy on all scam employers in the real world. The check on the as-collected feature table (Table 6.8) was made for this reason, and it supports the same ordering of the results.

7.1.2 The sources are not equally useful, and the registry is the weakest (RQ1, RQ3)

The AUC of each source alone (Table 6.4) shows a clear order: the model score 0.973, reputation 0.911, registry 0.661 and website 0.607. Two explanations can be given. First, registration shows that an organization exists, and it does not show that the organization is honest. All 23 fraudulent employers that the proposed fusion misses are registered, and the model gave them a mean score of only 0.47, so the registry evidence raised their score above the High boundary. Second, many genuine

small companies in Sri Lanka are not listed in the registers that can be searched automatically, so a missing entry is a weak sign. The 16 genuine employers that the Random Forest flags have no registry entry, no presence on a review platform, and one in four lacks HTTPS; they look like fraud on every layer. The design decision of Chapter 4 to treat a missing entry as “not confirmed” and not as “fake” is supported by these cases.

The registry still has a role that the AUC does not show. It is the only source that a job seeker can check independently on an official portal, and it is the only evidence that gives a positive proof of existence. The fusion without the registry has a higher AUC (0.985) than the full fusion (0.951), which means that on this data the fixed registry weight costs some ranking quality. The weight of 0.30 was kept because of the value of the evidence to the user and not because of its statistical power, and this is a trade-off that the thesis states openly.

7.1.3 Explainability has a measurable price (RQ3)

The proposed weights (0.40, 0.30, 0.20, 0.10) reach a fraud recall of 0.781 on the test split. The grid-search weights reach 0.933 and the model alone 0.914, at a similar false positive rate. The price of keeping the registry and the reputation layer visible in the score is therefore 13 to 15 points of fraud recall. The profile with a dominant model weight (0.876) shows that part of this price can be recovered without removing the other sources. The test accuracy of the proposed weights (0.870) is higher than that of equal weights (0.850), but the intervals overlap widely, and hypothesis H1 is supported in direction only (Table 6.9). The hypothesis H2 is clearly supported: a profile that is dominated by the registry, or that ignores reputation, loses more than half of the fraud (recall 0.429) and its interval does not overlap that of the proposed weights.

These findings answer the question that the literature leaves open (Section 2.5.1). The weights of a fusion score should not be set by assumption, because a plausible choice such as a registry-heavy profile fails, and they should not be set only to maximize a metric, because that removes evidence that the user can check. A justified middle position is possible, and its cost can be stated in numbers.

7.1.4 Bands turn the score into a safe decision rule (RQ4)

With the boundaries 45 and 70, no fraudulent employer falls in the Low band (0 of 107; upper 95% limit about 3.5%), 82 of 105 fall in High and 23 in Medium. Mean scores are 67.5 for genuine and 31.8 for fraudulent employers. The cost of this safety is that 72 genuine employers are placed in Medium and 16 in High. A cost-based analysis on the validation split gives a single cut-off of 65 for all three cost ratios (3:1, 5:1 and 10:1), which shows that the ranking of the cut-offs is not sensitive to the exact cost assumed. The alternative boundaries of 55 and 60 catch 91.4% of the fraud but let 5 fraudulent employers into Low, and Chapter 3 explained why a Low band without fraud was preferred. The Medium band is the price of that choice, and its meaning for the platform is discussed in Section 7.5.

7.2 Comparison with literature baselines and existing solutions

A direct numerical comparison with earlier work is not possible, because the studies use other countries, other registers and other data, and part of the data of this research was constructed. The comparison below is therefore qualitative and is made on method.

Table 7.1 compares this research with the closest studies and with the manual practice of a job seeker.

Table 7.1: Comparison with the closest studies

Study	What it did	What this research adds
Mahbub et al. [1]	Registry and website features raised a Random Forest from 89.22% to 91.86% accuracy (Australian jobs)	Sri Lankan registers, reputation evidence, a test of the combination weights on separate data, and risk bands. The accuracy of 91.7% is of the same order, but the datasets differ and no superiority is claimed
Alghamdi and Alharby [2]	Employer metadata among the strongest indicators	Evidence that can be checked by the user, with a report of what each source returned
Pangare et al. [3]	Hybrid checks (GST, CEO identity, Glassdoor) with text classification	Local identifiers instead of Indian ones; weights chosen by experiment and not by assumption; the cost of explainability reported
Registry lookup, review platforms, manual search (Table 2.3)	Each source is checked by hand, one at a time	One request combines the sources, gives a band and keeps links to the official source for each statement

Two points of the comparison are worth stating. The accuracy values are similar, which is expected because all these studies use a Random Forest on structured features. The difference lies in what is reported around the classifier: the contribution of each source, the way the weights were chosen, the price paid for explainability, and the thresholds. These are the items that the literature review identified as missing (Section 2.5.1). Compared with the manual practice of a job seeker, the component does not find new information; it collects in seconds the same public sources and states the result as a risk with reasons.

7.3 Research contribution

- A tested procedure for choosing fusion weights: the weights are compared on a validation split, the chosen profile is evaluated once on a test split grouped by domain, and the uncertainty is shown with Wilson and bootstrap intervals.
- A measured price of explainability: 13 to 15 points of fraud recall between the proposed weights and the best weights for this data, with the finding that registry-heavy and reputation-ignoring profiles are unsafe.
- A banding rule with a stated purpose: boundaries 45 and 70 that keep fraud out of the Low band on the held-out data, together with the cost of that rule (72 genuine employers in Medium).
- Evidence about Sri Lankan sources: the registers that can be checked automatically give a weak statistical signal, whereas reputation evidence is strong, and registered employers can still be fraudulent.
- A documented dataset construction, including its limits (Section 3.2.2), so that the results can be repeated and judged.

7.4 Product contribution

The product is a working service that takes a company name and website and returns a legitimacy score, a risk band, a confidence flag and an evidence report with the points of each layer and the official source of each statement. It queries seven registries and sources in parallel with the reputation and website checks, and degrades to a lower-confidence result when the website cannot be scanned. It is covered by 38 automated

tests and eight scenario tests (Chapter 5). In the complete solution, it has two uses: any user can check a company before applying, and a company that registers on the platform is checked automatically, with Low and Medium companies verified and High companies sent to an administrator with the evidence report. It also supplies the employer risk $r = 1 - S/100$ to the recommendation component.

7.5 Limitations and threats to validity

7.5.1 Limitations of the data

The dataset has 1,514 employers, 978 genuine and 536 fraudulent. A part of it was generated: the feature values of the fraudulent records, and some flags of the genuine records, were redrawn at fixed rates chosen by the author. The results are a controlled evaluation of the method and not a measurement on real scam traffic. Fraud labels that come from news reports are strong, whereas labels that come from social-media comments are unverified reports and may be wrong or malicious. The test split has 300 employers, 105 of them fraudulent, so the intervals are wide: the fraud recall of the proposed weights (0.781) has a Wilson interval of roughly 0.69 to 0.85. A single split with one random seed was used, and a repetition with other seeds or a cross-validation would show how much the results move.

7.5.2 Limitations of the method

The weights, the bands and the SME rule are fixed. A scammer who knows them can register a company, create a clean website and obtain reviews, and the 23 registered frauds that the fusion misses show that this already happens in the data. The evaluation used the normalized formula of Section 4.4.5, while the deployed scorer uses capped points with graded registry points and the SME rule. The 38 unit tests check that the deployed scorer follows its written rules, but the deployed scorer was not evaluated separately on the held-out data, so Tables 6.2 to 6.7 apply strictly to the evaluated formula, and the routing numbers of Section 6.4 assume the same band boundaries. The model score is a classifier output and has not been calibrated, and it should not be read as a probability.

7.5.3 Automatic verification of the Medium band

The registration flow verifies Low and Medium companies automatically and sends only High companies to an administrator. On the test split this stops 82 of 105 fraudulent employers (78.1%) and sends 16 of 195 genuine ones (8.2%) to an administrator. It also lets 23 fraudulent employers pass, all of them from the Medium band, which is 22% of the fraud and 11% of the 202 employers verified automatically. The Low band is clean, so the risk lies entirely in Medium, which is also the band where 72 genuine employers sit. The choice to verify Medium automatically reduces the work of the administrator, but it moves the cost of the model's errors to the job seekers who trust a verified company.

The component reduces this risk in three ways. The Medium band states that registration is not confirmed and asks the job seeker to check it with the employer. The evidence report names the weak points. The score is passed to the recommendation component, which ranks a Medium employer below a Low one. These measures help only if the platform shows the band and the report to the job seeker next to a verified company. A safer arrangement, recommended in Section 8.4, is to verify Medium companies provisionally, to show the "not confirmed" note, to keep the report with the registration and to review these companies later or sample them.

7.5.4 Limitations of the implementation and of the validation

- Defects D1 to D3 of Section 5.4.4 were found and not corrected in the evaluated version. D1 (an empty request receives a Medium band) matters most for the registration flow, because such a request would be verified automatically.
- The live registers and review platforms can change their pages, block automated access or time out. The service then returns a result with lower confidence, but the live behavior was not measured in the tests of Chapter 5, which replaced the sources by fixed inputs.
- No study with job seekers or administrators was carried out (Section 6.3), and no complete run through the mobile application and the recommendation component was made (Section 6.4). It is not known whether users understand the evidence report.
- The source code contains a default value for the CSE access key, and the library versions are not fixed in the requirements file. Both should be corrected before a public release (Section 8.4).

- The scope is Sri Lankan employers that publish information in English, and the component does not examine job advertisements or conversations.

7.5.5 Threats to validity

Internal validity is threatened by the construction of the data and by the choice of the weights on the same dataset family; the grouped split, the use of the validation split for tuning, and the check on the as-collected table reduce but do not remove this threat. Construct validity is threatened because the label “fraud” comes from public reports, and a company can be reported by mistake. External validity is limited by the size of the data, by the country and by the generated part. Statistical conclusion validity is limited by the size of the test split, and the intervals are reported for this reason.

7.6 Implications

For job seekers, the result is a way to see the reasons for a risk and to verify each of them with an official source, which is more useful than a bare label. The finding that a missing registry entry is a weak sign means that an unregistered small company should be asked for its registration number and not rejected. For platform operators, the bands give a practical rule for the work of administrators, and the evidence of this thesis shows that the Low band can be trusted on the data tested, whereas the Medium band needs a visible “not confirmed” status and a later review. For researchers, the thesis shows that the choice of weights and thresholds in a fusion score can and should be tested on separate data and reported with intervals, and that the price of explainability can be stated in numbers. For the public registers, the need to read pages automatically with a browser fallback shows the value of a published interface to the registers of companies, which would make the registry evidence faster and more reliable.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

Job seekers need to know whether an employer is real before they apply, but the evidence is spread over official registers, review platforms and the employer's own website, and it is slow to check by hand. This research designed, built and evaluated a component that collects this evidence for a Sri Lankan employer, combines it into a legitimacy score from 0 to 100, assigns a Low, Medium or High risk band, and explains the result with a report that links each statement to its source.

The evaluation on 1,514 labelled employers, split by website domain into training, validation and test sets, led to four main conclusions. First, a Random Forest on 14 structured features separates genuine from fraudulent employers well on unseen data (test accuracy 0.917, fraud recall 0.914, AUC 0.973). Second, no single source is enough: reputation evidence is strong (AUC 0.911), whereas the registry (0.661) and the website (0.607) are weak alone, and weight profiles dominated by the registry lose more than half of the fraud. Third, keeping the evidence visible in a fixed-weight score costs 13 to 15 points of fraud recall compared with the best weights, and this cost can now be stated. Fourth, with the boundaries 45 and 70 no fraudulent employer reached the Low band on the test split (0 of 107), at the price of 72 genuine employers in the Medium band.

The component is implemented as a service with 38 automated tests, and it is used in the complete solution to answer public checks and to screen company registrations, where Low and Medium companies are verified and High companies go to an administrator with the evidence report. The work also shows its limits: the data were partly generated, the Medium band lets 22% of the fraud pass when it is verified automatically, three defects were found in the service, and no user study was carried out. These limits define the future work of Section 8.4.

8.2 Achievement of objectives

Table 8.1 states, for each objective of Section 1.5.2, whether it was achieved and where the evidence is.

Table 8.1: Achievement of the objectives of Section 1.5.2

Objective	Status	Evidence
O1 Review related work and identify the gaps	Achieved	Chapter 2; gaps G1 to G5; the comparison in Table 7.1
O2 Multi-source evidence pipeline	Achieved	Registry, reputation and website layers with 14 features; seven registry sources; Chapters 4 and 5
O3 Random Forest on employer features	Achieved	Test accuracy 0.917, precision 0.952, recall 0.918, F1 0.935, AUC 0.973 (Table 6.1)
O4 Fusion weights found by experiment	Achieved	Five profiles and a grid search compared; price of explainability: 13 to 15 points of fraud recall (Tables 6.2 and 6.3)
O5 Risk bands that keep fraud out of Low	Achieved on the test split	0 of 107 in Low; 82 of 105 frauds in High; 72 genuine employers in Medium (Table 6.6)
O6 Service and integration	Partly achieved	38 of 38 automated tests; 6 of 8 scenario tests; interface and routing checked; no end-to-end run with the other components and three defects open (Sections 5.4 and 6.4)

The research questions are answered as follows. RQ1: Fourteen features from three layers can be collected automatically. RQ2: The Random Forest separates the classes with an AUC of 0.973 on unseen employers. RQ3: The proposed weights give a fraud recall of 0.781 against 0.933 for the best weights, and weights dominated by the registry fail. RQ4: The boundaries 45 and 70 keep the Low band free of fraud on the test split and send uncertain cases to the Medium band.

8.3 Individual research and product contributions

The author’s contribution to the research is the question of how the weights and thresholds of an explainable fusion score should be chosen and what that choice costs, with the dataset, the domain-grouped split, the weight and threshold experiments, the interval estimates and the error analysis that answer it. The contribution to the product is the employer legitimacy service: the registry, reputation and website layers; the Random Forest; the scoring layer with its evidence report; the SME rule that avoids penalizing small genuine companies for a missing registry entry; the automated tests; and the interface that the public check, the registration check and the recommendation

component use. The contribution to the group solution is the employer risk r = 1 − S/100 that is combined with the risks of the post detector and the conversation detector when the recommendation component ranks jobs.

8.4 Future directions

8.4.1 Corrections to the present version

- Return no band for an Unknown result and send such a request to the administrator (defect D1); catch errors around each parallel source call (D2); label or remove the field prediction (D3).
- Remove the default access key from the source code, read all keys from the environment, fix the library versions in the requirements file, and save the model with the version of scikit-learn that will run it.
- Evaluate the deployed scorer, with capped points and the SME rule, on the held-out split, so that the results apply to the code that is run.

8.4.2 Handling of the Medium band

- Verify Medium companies provisionally, show the “not confirmed” note to job seekers, and keep the evidence report with the registration so that a later review does not need a new check.
- Review Medium companies in a lower-priority queue or by sampling, and measure how many are later found to be fraudulent.
- Re-check verified companies at intervals, because a company’s website, reviews and scam reports change after registration.

8.4.3 Data and method

- Collect a larger set of real employers over time, with labels confirmed by authorities or by repeated reports, and replace the generated part of the data. Repeat the evaluation with several seeds or a cross-validation grouped by domain.
- Learn the weights and thresholds with cross-validation and a limit on how far the explainable weights may move from the proposed ones, and calibrate the model score so that it can be read as a probability.

- Add signals that are harder to fake than a website, such as the age and ownership history of the domain, the consistency of contact details across sources, and the history of the company in the register.

8.4.4 Validation and integration

- Run a complete test with the mobile application and the recommendation component on real job advertisements, and measure the response time of live checks.
- Carry out a study with job seekers on whether the evidence report is understood and changes their decision, and with administrators on the usefulness of the High-band cases, after ethical clearance.
- Seek a published interface to the company registers, so that the registry evidence no longer depends on reading web pages, and extend the component to Sinhala and Tamil sources.

REFERENCES

[1] S. Mahbub, E. Pardede, and A. S. M. Kayes, “Online recruitment fraud detection: A study on contextual features in Australian job industries,” IEEE Access, vol. 10, pp. 82776–82787, 2022.

[2] B. Alghamdi and F. Alharby, “An intelligent model for online recruitment fraud detection,” J. Inf. Secur., vol. 10, no. 3, pp. 155–176, 2019.

[3] T. Pangare, S. Surve, K. Salokhe, and S. Hasbe, “JobScamShield – Fake job post detection,” Int. J. Multidiscip. Res. (IJFMR), vol. 8, no. 2, Mar.–Apr. 2026.

[4] P. Probst, M. N. Wright, and A.-L. Boulesteix, “Hyperparameters and tuning strategies for random forest,” WIREs Data Min. Knowl. Discov., vol. 9, no. 3, art. e1301, 2019.

[5] M. H. Shaker and E. Hüllermeier, “Random Forest calibration,” arXiv preprint arXiv:2501.16756, 2025.

[6] O. Sagi and L. Rokach, “Ensemble learning: A survey,” WIREs Data Min. Knowl. Discov., vol. 8, no. 4, art. e1249, 2018.

[7] A. Fernández, S. García, M. Galar, R. C. Prati, B. Krawczyk, and F. Herrera, Learning from Imbalanced Data Sets. Cham, Switzerland: Springer, 2018.

[8] H. Guo, Y. Li, J. Shang, M. Gu, Y. Huang, and B. Gong, “Learning from class-imbalanced data: Review of methods and applications,” Expert Syst. Appl., vol. 73, pp. 220–239, 2017.

[9] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 30, 2017.

[10] M. T. Ribeiro, S. Singh, and C. Guestrin, ““Why should I trust you?”: Explaining the predictions of any classifier,” in Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining, San Francisco, CA, USA, 2016, pp. 1135–1144.

[11] K. F. Y. Lam, V. Gopal, and J. Qian, “Confidence intervals for the F1 score: A comparison of four methods,” arXiv preprint arXiv:2309.14621, 2023.

[12] scikit-learn developers, “scikit-learn: Machine learning in Python, user guide,” 2026. [Online]. Available: https://scikit-learn.org/stable/

[13] OpenAPI Initiative, “OpenAPI Specification, version 3.1.0,” 2021. [Online]. Available: https://spec.openapis.org/oas/v3.1.0

[14] Parliament of the Democratic Socialist Republic of Sri Lanka, Personal Data Protection Act, No. 9 of 2022, Colombo, Sri Lanka, 2022.

[15] Association for Computing Machinery, “ACM Code of Ethics and Professional Conduct,” 2018. [Online]. Available: https://www.acm.org/code-of-ethics

[16] S. Ramírez, “FastAPI documentation,” 2026. [Online]. Available: https://fastapi.tiangolo.com

[17] S. Vidros, C. Kolias, G. Kambourakis, and L. Akoglu, “Automatic detection of online recruitment frauds: Characteristics, methods, and a public dataset,” Future Internet, vol. 9, no. 1, art. 6, 2017.

GLOSSARY

Term	Definition
Auto-verification	Acceptance of a company registration by the platform without a person deciding. In this thesis, it applies to the Low and Medium bands.
AUC	Area under the ROC curve. The probability that a randomly chosen genuine employer receives a higher score than a randomly chosen fraudulent one; 0.5 is chance and 1.0 is perfect ranking.
Band (risk band)	One of three ranges of the legitimacy score: Low risk (70 or more), Medium risk (45 to 69) and High risk (below 45).
Bootstrap interval	An interval for a statistic obtained by recomputing it on many samples drawn with replacement from the test data.
Calibration	The property that a score of 0.8 corresponds to a true share of about 80%. The Random Forest score of this work is not calibrated.
Confidence flag	A field of the response that says high or low confidence, set low when the website could not be scanned or too little information was given.
Cost ratio	The relative cost of a missed fraud compared with a false alarm (3:1, 5:1 and 10:1 in this work), used to choose a threshold.
Domain-grouped split	A division of the data in which all records of one website domain fall in the same part, so that the test set holds no near-duplicate of a training record.
Employer legitimacy	The degree to which an employer is a real, registered organization that offers real jobs, as judged from public evidence.
Evidence reports	The part of the response that lists what each source returned, the points of each layer and the reasons for the band, with the source of each statement.
False positive rate (FPR)	The share of genuine employers that are wrongly flagged as fraudulent.
Fraud recall	The share of fraudulent employers that are caught, that is, placed in the fraudulent class or the High band.
Fusion	The combination of the model score and the scores of the registry, reputation and website layers into one legitimacy score by a weighted sum.
Held-out data	Data that was not used to train the model or to choose the weights and thresholds.
Legitimacy score	A number from 0 to 100 where a higher value means a more trustworthy employer; written S_B in the fusion formula and S elsewhere.
Model score	The average class probability of the trees of the Random Forest for the class legitimate; not a calibrated probability.
Random Forest	A classifier made of many decision trees trained on random samples of the data and features, whose votes are averaged.
Registry layer	The part of the service that looks for the company in official registers, such as the Colombo Stock Exchange, the Department of Registrar of Companies (eROC) and the Board of Investment.
Reputation layer	The part of the service that looks for the company on review and professional platforms and for scam reports.
SME rule	A rule of the scoring layer that lifts the score of a small company to 48 when it has a clean footprint but no registry entry, so that a missing entry alone does not place it in the High band.
Threshold	A score value above which an employer is treated as legitimate; the cut-off for the model is 0.5 and for the legitimacy score 45 and 70.

Website layer	The part of the service that reads the company website and checks HTTPS, About and Contact pages, privacy terms and signs of weak security.
Wilson interval	A confidence interval for a proportion that stays valid for small counts and for proportions near 0 or 1.

APPENDICES

Appendix A Extended results and technical specification

This appendix gathers material that supports Chapters 4 to 6 and is too detailed for the main text.

A.1 Settings used

Table A.1 lists the settings of the model, the fusion and the bands.

Table A.0.1: Settings of the model, the fusion and the bands

Item	Value
Random Forest	200 trees; maximum depth 8; minimum 16 samples to split; minimum 8 samples per leaf; balanced class weights; seed 42
Features	14 structured features from the registry, reputation and website layers
Data	1,514 employers (978 genuine, 536 fraudulent); training 901, validation 313, test 300 (195 genuine, 105 fraudulent); split grouped by website domain
Evaluated fusion	S = 100 × (0.40 × model + 0.30 × registry + 0.20 × reputation + 0.10 × website)
Risk bands	Low 70 or more; Medium 45 to 69; High below 45
Deployed scorer	Capped points: model 40, registry 30, reputation 20, website 10; registry points CSE 30, BOI/CBSL/IRCSL/SLAASMB 25, DRC 20, other government 15, unverified 10; SME floor to 48
Cost ratios	3:1, 5:1 and 10:1 (missed fraud: false alarm); threshold sweep 20 to 80 in steps of 5

A.2 Detailed test results

Table A.2 gives the fraud recall of each weight profile, Table A.3 the AUC of each source and combination, and Table A.4 the risk bands for two choices of boundaries, all on the test split.

Table A.2: Fraud recall of the weight profiles on the test split

Profile	Fraud recall
Grid search (0.50, 0.10, 0.25, 0.15)	0.933
Model alone (Random Forest, cut-off 0.5)	0.914
Model-dominant	0.876
Proposed explainable weights (0.40, 0.30, 0.20, 0.10)	0.781
Equal weights	0.743
Registry-heavy	0.429
Reputation ignored	0.429

Table A.3: AUC on the test split by source and by combination

Source or combination	AUC
Model score	0.973
Fusion without the registry	0.985
Explainable fusion	0.951
Reputation	0.911
Fusion without the model	0.873
Registry	0.661
Website	0.607

Table A.4: Risk bands on the test split (genuine/fraudulent employers)

Bands	High	Medium	Low	Fraud in Low
45 and 70 (deployed)	16 / 82	72 / 23	107 / 0	0
55 and 60	29 / 96	13 / 4	153 / 5	5

A.3 Response fields of the service

The POST /predict response contains: prediction, probability, confidence and warning (the model part and the confidence flag); risk_score, risk_level, verdict, color and recommendation (the score, band and advice); evidence and score_breakdown (the evidence report and the points of each layer); registration_status, registration_status_label, registration_summary, registration_trace, registration_sources, registration_method, government_registration_source, reg_name, reg_number, opencorporates_url and slaasmb_sbe_name (the registry result); features_used (the 14 features and the signals); and elapsed_seconds (the time taken).

A.4 Automated tests and scenario tests

The 38 automated tests are in the files tests/test_scoring_layer.py (31 tests) and tests/test_model.py (7 tests) in the service folder, and are run with python -m pytest tests. The scenario tests of Table 5.4 call the HTTP interface with the registry, review and website calls replaced by fixed inputs. The inputs were: (1) empty request; (2) company name with a TopJobs employer link; (3) a CSE-listed company with LinkedIn, Glassdoor, Facebook and review evidence; (4) an unlisted private limited company with a complete site, a LinkedIn page and a Facebook page; (5) the same company with a site that returns no content; (6) a company with no registry match, a

scam report, payment wording and a thin site on a .xyz domain; (7) the same company as in (4) with the registry call raising an error; (8) 20 repeated requests as in (4).

A.5 Reproducing the results

The held-out results of Chapter 6 are regenerated by running python notebooks/threshold_analysis.py from the service folder with the seed 42. The service is started with run.ps1 and listens on port 8001.

Appendix B Academic integrity and AI use disclosure

This appendix states how artificial intelligence tools were used in preparing this dissertation, in line with the academic integrity policy of the module.

Table B.1 lists the tools, their purpose, the parts affected and the verification performed.

Table B.1: Use of AI tools

Tool	**Purpose**	**Parts affected**	**Verification**
Claude (Anthropic)	Drafting and editing text from my results; writing the automated tests	Chapters 5 to 8, back matter, tests	Numbers checked against project outputs; tests run; text reviewed and edited by me

The research design, the dataset, the model, the service and the experiments are the work of the author. The AI assistant did not collect or label data and did not run any experiment that is reported as a research result; the results of Chapter 6 come from the project's own scripts. The author takes responsibility for the content of this dissertation, including any text produced with the help of the tools above.